

Title: Durably reducing conspiracy beliefs through dialogues with AI

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Abstract: Conspiracy theory beliefs are notoriously persistent. Influential theories propose they fulfill important psychological needs, thus resisting counterevidence. Yet previous failures in correcting conspiracy beliefs may be due to counterevidence being insufficiently compelling and tailored. To evaluate this possibility, we leverage developments in generative artificial intelligence and engaged 2,190 conspiracy believers in personalized evidence-based dialogues with GPT-4 Turbo. The intervention reduced conspiracy belief by ~20%. The effect remained 2 months later, generalized across a wide range of conspiracy theories, and occurred even among participants with deeply entrenched beliefs. Although the dialogues focused on a single conspiracy, they nonetheless diminished belief in unrelated conspiracies and shifted conspiracy-related behavioral intentions. These findings suggest that many conspiracy theory believers can revise their views if presented with sufficiently compelling evidence.

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Main Text: Widespread belief in unsubstantiated or false conspiracy theories is both a major source of public concern and focus of scholarly research (1–3). Conspiracy theories – in which events are understood as being caused by secret, malevolent plots involving powerful conspirators – are often quite implausible, yet a large fraction of the world has come to believe them, including as much as 50% of the US population by past estimates (4–7). Such prevalence is particularly concerning, given that conspiracy belief is often used as a paradigmatic example of resistance to evidence (8–10): there is little evidence of interventions that successfully debunk conspiracies among people who already believe them (11, 12).

The apparent resilience of conspiracy theories in the face of clear counter-evidence poses a powerful challenge to scientific theories that emphasize the role of reasoning in belief formation and revision (13, 14). Instead, belief in conspiracies has primarily been explained through social-psychological processes thought to blunt rational decision-making and receptivity to evidence (7, 15–19). Popular explanations propose that people adopt conspiracy theories to sate underlying psychic “needs” or motivations, such as the desire for control over one’s environment and experiences (15), certainty and predictability (20), security and stability (21), and uniqueness (22). If these psychological needs are met by believing in conspiracy theories, the beliefs become more than just opinions; they become mechanisms for psychological equilibrium – and thus are argued to be highly resistant to counterevidence (1, 3, 23). Coupled with peoples’ motivations to maintain their identity and/or group memberships, with which conspiracies also interface (24–26), believers may use specific forms of biased information processing (motivated reasoning) where counterevidence is selectively ignored (27–29).

These perspectives, which center the psychological drives of those who believe in conspiracies, paint a grim picture for countering conspiratorial beliefs: Because conspiracy believers at some level “want” to believe, convincing them to abandon unfounded beliefs using facts should be virtually impossible (without more fundamentally altering their underlying psychology and identity commitments).

Here, we question this conventional wisdom about conspiracy theories and ask whether it may, in fact, be possible to talk people out of the conspiratorial “rabbit hole” with sufficiently compelling evidence. Leveraging recent advancements in Large Language Models (LLMs), we shed new light on whether counterevidence reduces belief in conspiracy theories. We hypothesize that fact-based interventions may appear to fall short due to a lack of depth and personalization of the corrective information. Entrenched conspiracy theorists are often quite knowledgeable about their conspiracy of interest, deploying prodigious (albeit often erroneous or misinterpreted) lists of evidence in support of the conspiracy that can leave skeptics outmatched in debates and arguments (30, 31). Furthermore, people believe a wide range of conspiracies, and the specific evidence brought to bear in support of even a particular conspiracy theory may differ substantially from believer to believer. Canned debunking attempts that argue broadly against a given conspiracy theory may, therefore, be ineffective because they fail to address the specific evidence held by the believer – and thus fail to be convincing.

In contrast, we hypothesize that LLMs offer a promising solution to these challenges because they possess two key capabilities: (i) access to a vast amount of information across diverse topics, and (ii) the ability to tailor counterarguments to specific conspiracies, reasoning, and evidence the believer brings to bear (32). These capabilities allow LLMs to respond directly to –

and refute – the particular evidence supporting an individual’s conspiratorial beliefs. In so doing, LLMs can potentially overcome the heterogeneity in conspiracy beliefs and supporting evidence that we hypothesize have stymied previous debunking efforts.

To test whether LLMs can effectively refute conspiracy beliefs – or whether psychological needs and motivations render conspiracy believers impervious to counterevidence – we develop a pipeline for conducting behavioral science research using real-time, personalized interactions between research subjects and LLMs. In our experiments, participants articulate a conspiracy theory in which they believe – in their own words – along with the evidence they think supports the theory. They then engage in a back-and-forth interaction with an artificial intelligence (AI) implemented using the LLM GPT-4 Turbo (33). In line with our theorizing around the unique capacities of LLMs for debunking conspiracies, we prompt the AI to use its store of knowledge to try to respond to the specific evidence raised by the participant and reduce the participant’s belief in the conspiracy theory (or, in a control condition, participants converse with AI about an unrelated topic). In particular, the AI was instructed to “very effectively persuade” users against belief in their chosen conspiracy, allowing it to flexibly adapt its strategy based on the participant’s unique arguments and evidence. To further enhance this tailored approach, we provided the AI with each participant’s written conspiracy rationale as the conversation’s opening message, along with the participant’s initial rating of their belief in the conspiracy. This design choice directed the AI’s attention to refuting specific claims, while simulating a more natural dialogue where the participant had already articulated their viewpoint. For the full prompts given to the model, see Table S2. The conversation lasted 8.4 minutes on average and comprised three rounds of back-and-forth interaction (not counting the initial elicitation of reasons for belief from the participant), a length chosen to balance the need for substantive dialogue with pragmatic concerns around study length and participant engagement.

This design allows us to test whether tailored persuasive communication is indeed able to reduce already-held conspiracy beliefs; how the effectiveness of such communication varies based on factors such as the intensity of the preexisting belief, the participant’s subscription to a more general conspiratorial mindset, the importance of the conspiracy to the participant’s life, and the content of the specific conspiracy theory articulated by the participants; and whether any such persuasion spills over to other related beliefs and behaviors. Finally, our design produces rich textual data from thousands of conversations between the AI and the human participants (<https://8cz637-the.shinyapps.io/ConspiracyDebunkingConversations>), which we analyze to gain insight into what the humans believe and how the LLM engages in persuasion.

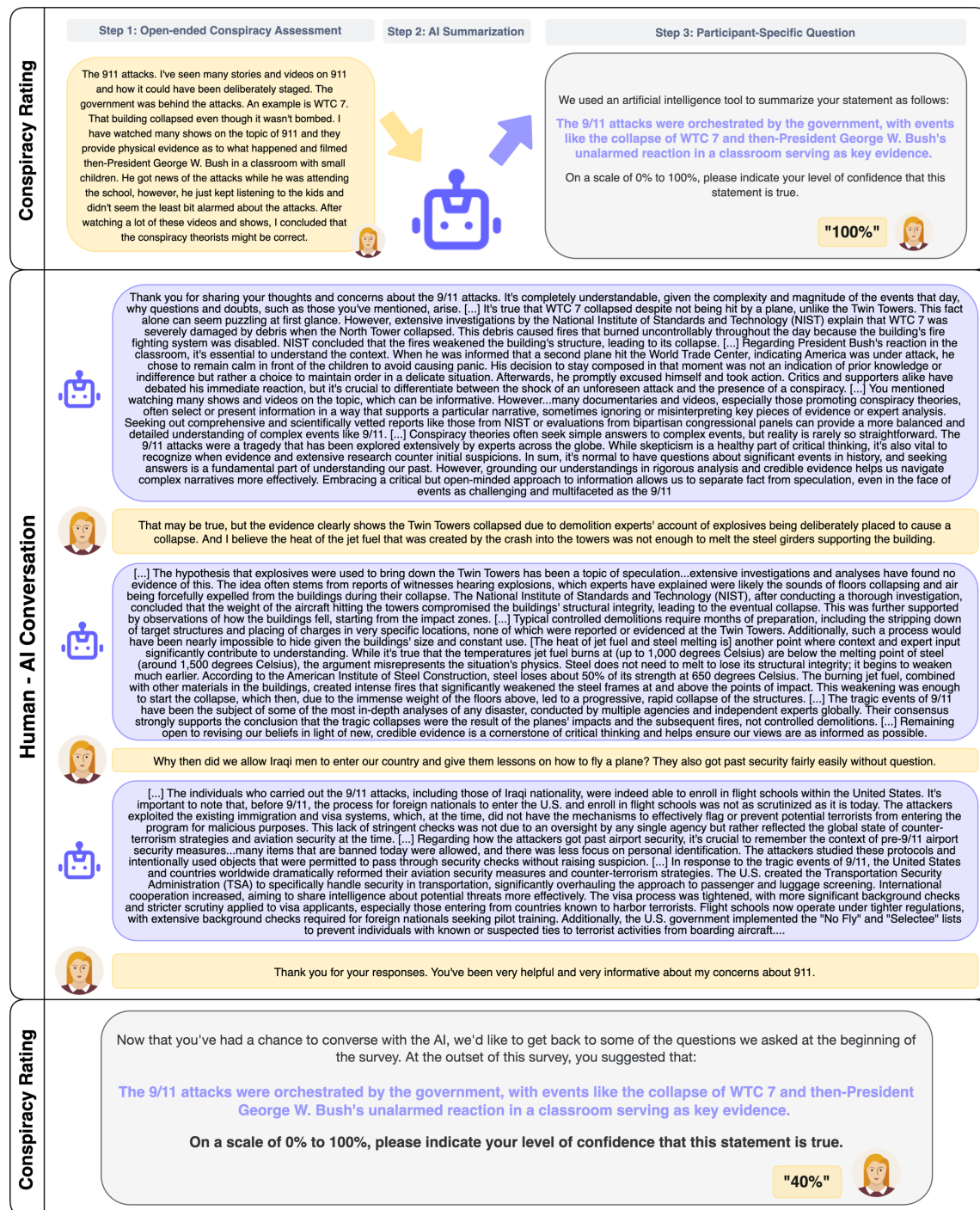


Figure 1. Design and flow of the human-AI dialogues. Respondents (yellow) described a conspiracy theory they believed in, along with the evidence they thought supported it. Each response was fed-forward to a query instructing the AI model (GPT-4 Turbo, shown in purple) to generate a brief, relatively standardized statement of that conspiracy. Participants then rated their belief in the summary statement, yielding our pre-treatment measure (0-100 scale, with 0 being “definitely false”, 50 being “uncertain” and 100 being “definitely true”). All respondents then entered into a conversation with the AI model (treatment argued against the conspiracy theory’s veracity, control discussed relevant topics). Following three rounds of dialogue, respondents once

again rated their belief in the summarized conspiracy statement, serving as our post-treatment measure. Shown is an example treatment dialogue which led the participant to substantially reduce their belief.

Can conspiracy beliefs be refuted?

In Study 1, participants indicated their belief in 15 popular conspiracy theories (from the Belief in Conspiracy Theories Index, BCTI), completed a distractor task, and were then asked to identify and describe a particular conspiracy theory they believed in (not necessarily one of the 15 rated earlier) as well as providing details about evidence or experiences supporting their belief. In real time, the AI created a summary statement of each participant's free-text conspiratorial belief description, and each participant was then asked to indicate their belief in the AI summary of their conspiracy statement – providing a pre-treatment measure of belief. This open-ended measurement approach avoids a longstanding criticism of discrete conspiracism measures, such as the BCTI, for failing to representatively sample from the universe of possible conspiracies (34).

Out of $N=1,055$ American participants (quota-matched to the U.S. census on age, gender, race, and ethnicity) who completed the pre-treatment measures, 72.2% indicated belief in a conspiracy theory and were included in our subsequent analyses, whereas 20.6% said they did not believe any conspiracy theories or described a belief that the AI classified as not actually conspiratorial (for coding validation, see Supplementary Materials, SM, section 1 and Table S4), 3.5% described a conspiracy theory but had belief below the scale midpoint, and 3.6% described a conspiracy theory that was inaccurately summarized by the AI.

To assess whether the AI could reduce conspiracy beliefs, participants were then randomly assigned to either have a 3-round conversation with the AI about their favored conspiracy belief (treatment group, 60% of the sample) or to participate in a similarly structured conversation about a neutral topic (control group, 40% of the sample). Although past work has typically found that people are less receptive to corrections (35), advice (36), and persuasion (37) labeled as coming from AI, we opted to avoid deception and explicitly informed participants that they were interacting with an AI.

For each participant, the AI was (a) provided with that participant's specific open-ended response, including their stated rationale for believing the conspiracy theory and their degree of endorsement and (b) prompted to use simple language to persuade the user that their conspiracy theory is not supported and change their beliefs to be less conspiratorial. Following the conversations, all participants re-rated belief in their stated conspiracy theory and the BCTI items (see **Figure 1** for key methodological steps and a sample conversation).

Was conversing with an AI able to successfully reduce participants' conspiratorial beliefs? Indeed, the treatment reduced participants' belief in their stated (i.e., focal) conspiracy by 16.8 points more than the control (linear regression with robust standard errors controlling for pre-treatment belief, 95% CI [13.8, 19.7], $p < .001$, $d = 1.15$; **Figure 2a** and SM section 2). This translates into a 21.43% decrease in belief among those in treatment (vs. 1.04% in the control). Furthermore, over a quarter (27.4%) of participants in the treatment became uncertain in their conspiracy belief (i.e. belief below the scale midpoint) following the conversation, compared to only 2.4% in the control. We also find a significant effect when examining the subset of focal conspiracies ($n = 303$) that strongly resembled BCTI items, using the pre- and post-treatment

BCTI ratings as opposed to the pre- and post-treatment evaluations of the AI summary ($b = -12.04$, 95% CI $[-16.63, -7.46]$, $p < .001$, $d = .70$; see SM Section 2.1), indicating the robustness of the results to our measurement approach.

To assess the persistence of this effect, we recontacted participants 10 days and 2 months later for a short follow-up in which they once again completed the outcome measures. We find no significant change in belief in the focal conspiracy theory from immediately after the AI conversation to either 10 days or 2 months later in a mixed-effects model with fixed effects for experimental condition and time point and random intercepts for participants ($b_{\Delta\text{ImmediatelyPost} - 10\text{Days}} = 0.63$, 95% CI $[-2.72, 1.46]$, $p = .56$; $b_{\Delta\text{ImmediatelyPost} - 2\text{Months}} = 0.03$, 95% CI $[-2.24, 2.31]$, $p = .98$; **Figure 2A** and Table S9). This result is robust to assuming that the 14% of participants who did not complete the follow-up returned to their initial pre-treatment belief levels ($b = 12.70$, 95% CI $[9.47, 15.93]$, $p < .001$). Thus, the change in beliefs we observe is remarkably persistent.

However, in Study 1, the proportion of participants who endorsed a conspiracy via free-text response was somewhat higher than prior estimates of the American public (4). Given that participants in Study 1 completed the BCTI before supplying their conspiracy theory, it is possible that exposure to the BCTI items increased the salience of particular conspiracy theories, and thereby increased reported belief.

We explore this possibility, as well as the replicability of our results and robustness to minor design changes, in Study 2 where $N=2,286$ Americans completed an extremely similar procedure without the BCTI. We also changed the wording for the conspiracy elicitation prompt such that, instead of directly asking participants which conspiracy theories they believed in, we provided a definition of what a conspiracy theory is and asked participants if they found any such theories compelling. Finally, we disabled copy-paste functionality to guard against participants themselves using LLMs to complete the study (38). Here, 64.6% of participants indicated belief in a conspiracy theory (see Table S3). Most importantly, we replicate the experimental results of Study 1. Participants in the treatment in Study 2 reduced belief in their focal conspiracy by 12.3 points more than participants in the control (95% CI $[10.07, 14.72]$, $p < .001$, $d = 0.79$; **Figure 2B** and Table S8), translating into a 19.41% average decrease in belief (versus a 2.94% decrease in the control).

Further demonstrating the robustness of our results, we also replicated our findings in a supplemental study conducted using a sample recruited through the participant supplier Lucid ($b = -10.99$, 95% CI $[-16.09, -5.88]$, $p < .001$, $d = .53$; see SM section 8 and Figure S13), which provides relatively inattentive respondents who mostly do non-academic surveys (39). Thus, the effect is not unique to attentive and engaged participants from Cloud Connect.

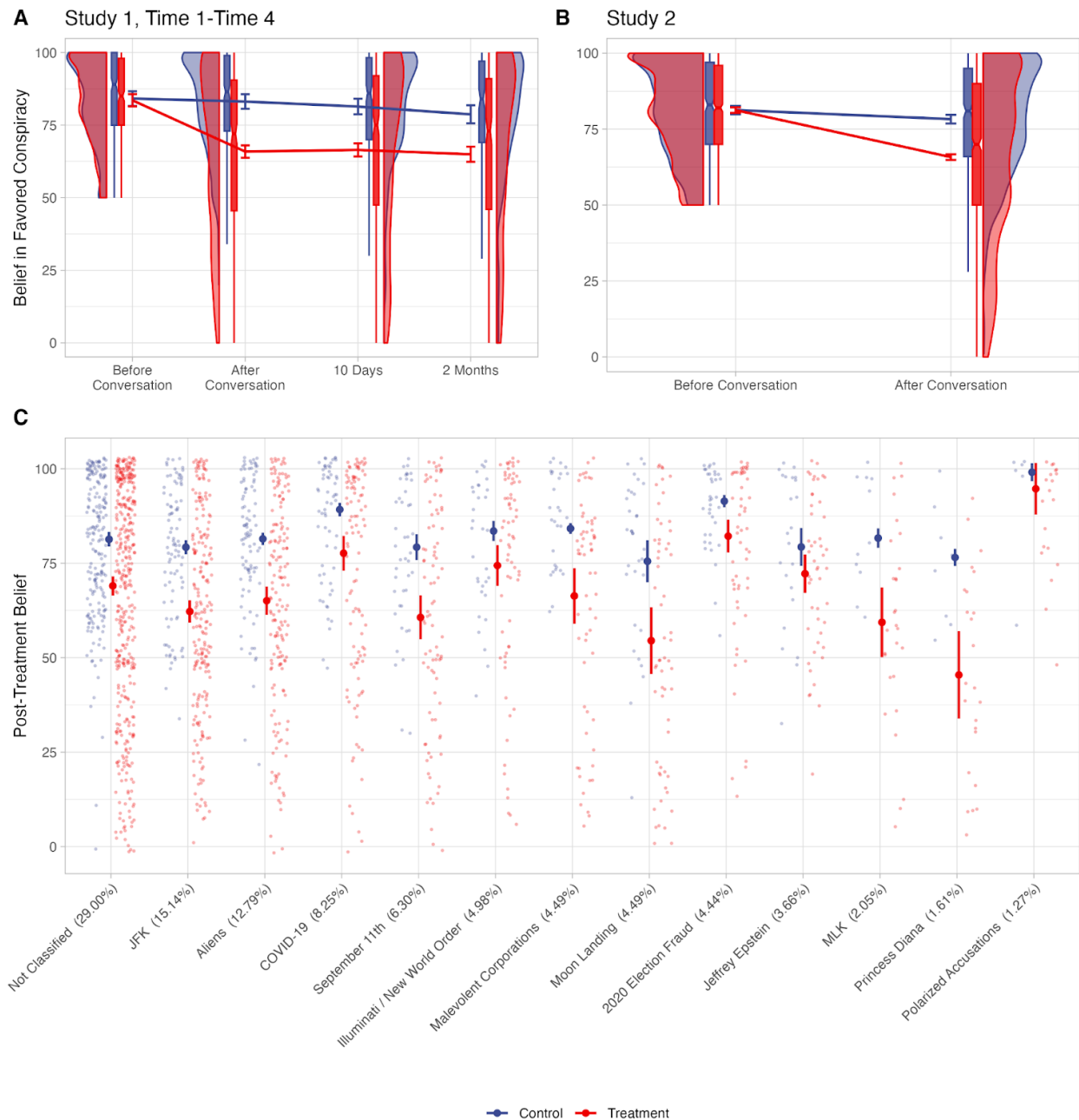


Figure 2. A brief conversation with an AI model durably reduces belief in conspiracy theories. Top: Average belief in each participant's focal conspiracy theory in by condition (treatment, in which the AI attempted to refute the conspiracy theory, in red; control, in which the AI discussed an irrelevant topic, in blue) and time point for Study 1 (A) and Study 2 (B). Before-conversation belief is greater than 50 for all participants because participants with initial belief below 50 were excluded from the study. Bottom: Belief immediately after the AI conversation by condition and topic of the participant's focal conspiracy theory; see SM section 3 for details of topic detection. Error bars indicate 95% confidence intervals.

Robustness across topics and people

Next, we examine the robustness of the AI conversation treatment effect. We begin by investigating whether the treatment size varies across the specific focal conspiracy theories articulated by the participants. To do so, we used a multi-step natural language processing and clustering approach to classify each focal conspiracy theory according to its contents (see SM section 3). We find that the treatment effect did not differ significantly across conspiracy type in an omnibus test ($F[12, 1971] = 1.30, p = .21$), and that the treatment significantly decreased belief across all but one of the 12 different types of conspiracy theory identified with $> 1\%$ prevalence in the sample (**Figure 2c**). Notably, the treatment worked even for highly salient – and likely deeply entrenched – political conspiracies such as those involving fraud in the 2020 US Presidential Election ($b = 10.61 [5.54, 15.67], p < .001, d = .82$) and the COVID-19 pandemic ($b = 11.79 [6.98, 16.60], p < .001, d = .73$). In addition to allowing us to test for the robustness of our treatment, this classification based on the participant’s open-ended responses also provides novel descriptive insight into which particular conspiracy theories Americans subscribe to. Alternative specifications of this clustering solution yielded highly similar patterns (see SM section 3.1; Figures S6-S7).

We now turn to variation in effect sizes across individuals. In particular, we ask whether the treatment is effective even among participants likely to have particularly entrenched beliefs. We use generalized additive models (GAMs) to analyze how the treatment effect varies in a non-linear manner based on several measures relevant to entrenchment. First, we examine participants’ level of pre-treatment belief in the focal conspiracy, and find that it does significantly moderate the treatment effect, resulting in a u-shaped curve ($\Delta AIC = -3.25, \Delta R^2 = .002, p = .022$; **Figure 3A** and Table S14). Second, we examine how important participants indicated the conspiracy theory is to their worldview (**Figure 3B** and Table S15), which does significantly decrease the size of the treatment effect ($\Delta AIC = 3.12, \Delta R^2 = .003, p = .025$). Critically, however, the effect was significant even among those who indicated the highest level of importance ($b = 5.84 [0.33, 11.35], p = .038, d = .53$). Third, we examined participants’ level of general conspiratorial ideation (i.e. the intensity with which they believed BCTI conspiracies), which showed non-significant moderation of the treatment effect ($\Delta AIC = 0.77, \Delta R^2 = .002, p = .108$; **Figure 3C** and Table S16). Participants at or above the 90th percentile of conspiratorial ideation in our sample (i.e., endorsing virtually all of the 15 diverse conspiracy statements) still displayed a substantial average treatment effect of $b = 9.07$ (95% CI $[2.73, 15.44], p = .006, d = .53$).

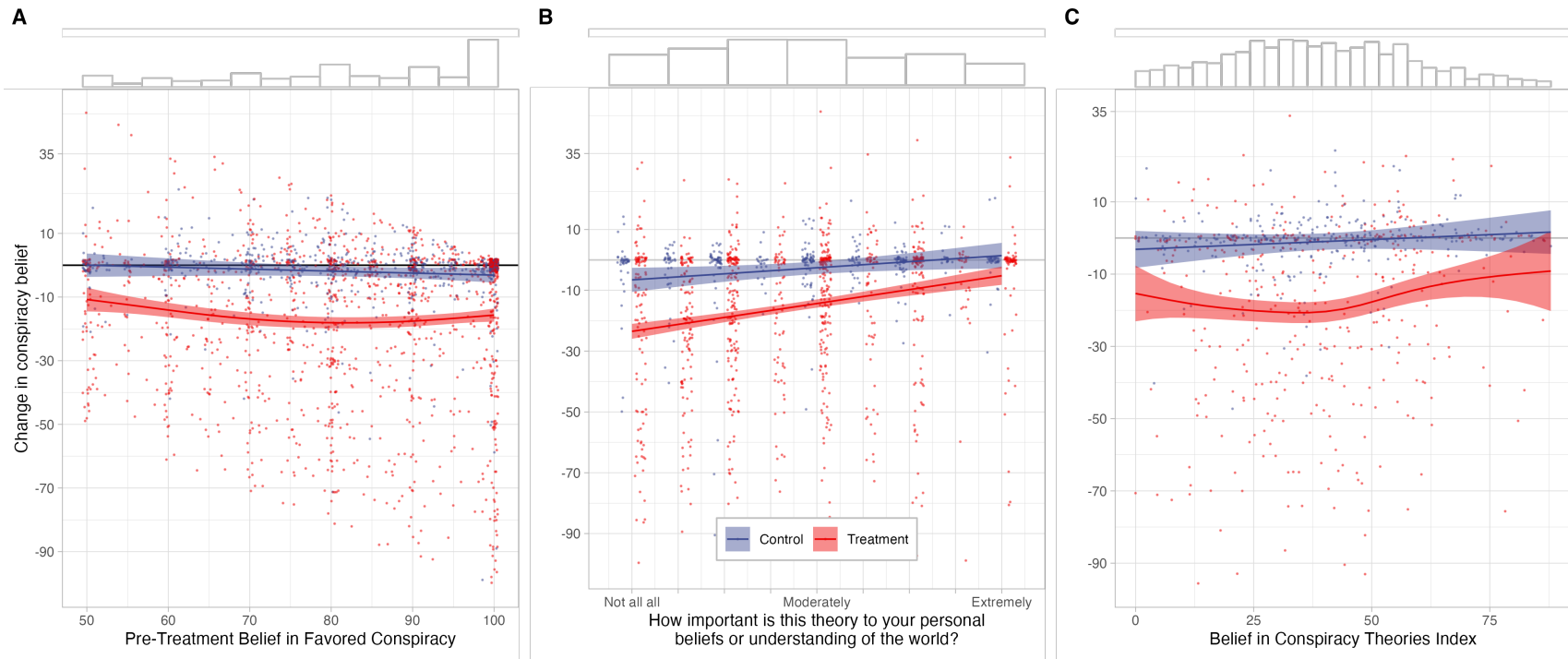


Figure 3. The treatment is effective even for those who are strongly attached to their conspiracy beliefs. Shown is the change in belief in the focal conspiracy from before AI conversation to after AI conversation, for the treatment (red) and control (blue) conditions. Data are pooled across studies to maximize power. Individual observations are plotted along with fit lines and 95% confidence intervals generated using generalized additive models. We conduct separate analyses for predictors of participant's pre-treatment level of belief in the focal conspiracy (A), rating of how important the focal conspiracy is to their personal beliefs or understanding of the world (B), and general conspiratorial mindset as measured by average belief in 15 conspiracies from the Belief in Conspiracy Theories Index completed pre-treatment (C).

We also examine moderation by demographic characteristics (age, race, gender, education) and other individual difference variables (political orientation, political extremism, religiosity, familiarity with generative AI, usage of generative AI, trust in generative AI, and institutional trust). In a single linear regression model including all candidate moderators and their interaction with experimental condition, as well as a control for conspiracy type and its interaction with experimental condition, only (a) trust in generative AI and (b) institutional trust consistently moderated the treatment effect, such that those higher in both kinds of trust showed larger treatment effects (see Tables S17-18). We conducted a post hoc analysis using the causal forest method (40) to further clarify and identify heterogeneous effects of the intervention across all moderators (including conspiracy type, pre-treatment beliefs, and importance) (see SM section 4.3). Variable importance analyses indicated that, for Experiment 1, the predominant determinants of treatment effect heterogeneity (in order) were participant's age, trust in generative AI, and BCTI scores; in Experiment 2, these were institutional trust (which was not measured in Experiment 1), trust in generative AI, age, and conspiracy-importance. Critically, while there were heterogeneous treatment effects across subgroups ($t = 4.97, p < .001$), the conditional average treatment effects (CATE) across covariate profile subgroups ranged from -20.54 to -6.56 – implying that the treatment reduced belief for all subgroups. For example, the CATE ranged from -17.7 to -4.5 (median = -9.7) for individuals who rated their focal conspiracy belief as “extremely important” to their personal beliefs; from -13.6 to -6.7 (median = -9.8) for individuals with minimal trust in AI; and from -18.2 to -10.0 (median = -15.4) for individuals with 95th percentile and above BCTI scores.

Spillover effects and behavioral implications

Next, we examined treatment effects on outcomes beyond belief in the focal conspiracy. First, we ask whether the treatment effect affected individuals' beliefs in conspiracy theories that were not discussed during the conversation with the AI model (see SM sections 2 and 7). We did so by analyzing respondents' belief in 15 widespread conspiracy theories from the BCTI (which is assessed by both pre-treatment and post-treatment in Study 1). We employed a linear mixed model with fixed effects for experimental conditions and time point (pre, post, 10-day, 2-months) and random intercepts for participant. Post-intervention, there was a 3.05-point decrease in general conspiracy beliefs in the active condition (95% CI [-3.90, -2.20], $p < .001$, 8.2% decrease; **Figure 4A** and Table S10), compared to a 1.64-point increase in the control ($d = .21$). This effect was still evident at the 2-month follow-up, with a 2.46-point decrease from pre-treatment (95% CI [-3.44, -1.49], $p < .001$). When only analyzing belief in BCTI conspiracy theories that a given participant believed pre-treatment (i.e. endorsed above the scale midpoint), the impact was more pronounced: a 9.39-point reduction immediately post-intervention (95% CI [-11.06, -7.72], $p < .001$, 12% decrease, **Figure 4B** and Table S11), compared to a 3.32-point reduction in the control ($d = 0.53$). This difference between treatment and control persisted at the 2-month follow-up ($b_{\Delta Treatment - Control} = -5.34$, 95% CI [-8.40, -2.29], $p < .001$).

In Study 2, we investigated the treatment's influence on participants' behavioral intentions (see SM section 5). We found that the treatment significantly increased intentions to ignore or unfollow social media accounts espousing the focal conspiracy ($\beta = .39$ [.27, .50], $p < .001$; **Figure 4C** and Table S19) and significantly increased willingness to ignore or argue against people who believe the focal conspiracy ($\beta = .42$ [.31, .54], $p < .001$; **Figure 4D** and Table S20). There was a directional but non-significant decrease in intentions to join pro-conspiracy protests ($\beta = -.12$ [-.27, .03], $p = .12$; **Figure 4E** and Table S21) – intentions which were low at baseline, potentially creating a floor effect.

How accurate is the AI?

Although it was not possible for us to ensure that all the claims produced by the AI in our experiment were accurate, we hired a professional fact-checker to evaluate the veracity and potential bias of all 128 claims made by GPT-4-turbo across representative example conversations from each of the 11 major conspiracy clusters generated by participants in our experiments. Of these claims, 127 (99.2%) were rated as “true”, 1 (0.8%) as “misleading” and 0 as “false”; and none of the claims were found to contain liberal or conservative bias. Together with a recent benchmarking study that found only 2.5% of the claims produced by GPT-4-turbo when summarizing text were hallucinations (41), these findings give us reason to believe that the information provided by the AI in our studies was largely accurate. Furthermore, in 1.2% of cases the participant named a focal conspiracy that is unambiguously true (e.g., MK Ultra). In these cases, the treatment effect was non-significant and directionally positive ($b = 6.51$, 95% CI [-39.42, 52.45], $p = .76$, $d = .43$), and significantly different from the effect for the other conspiracies ($b_{\Delta \text{True} - \text{False Conspiracies}} = -20.57$, 95% CI [-33.14, -8.00], $p = .001$; see SM Section 2.3).

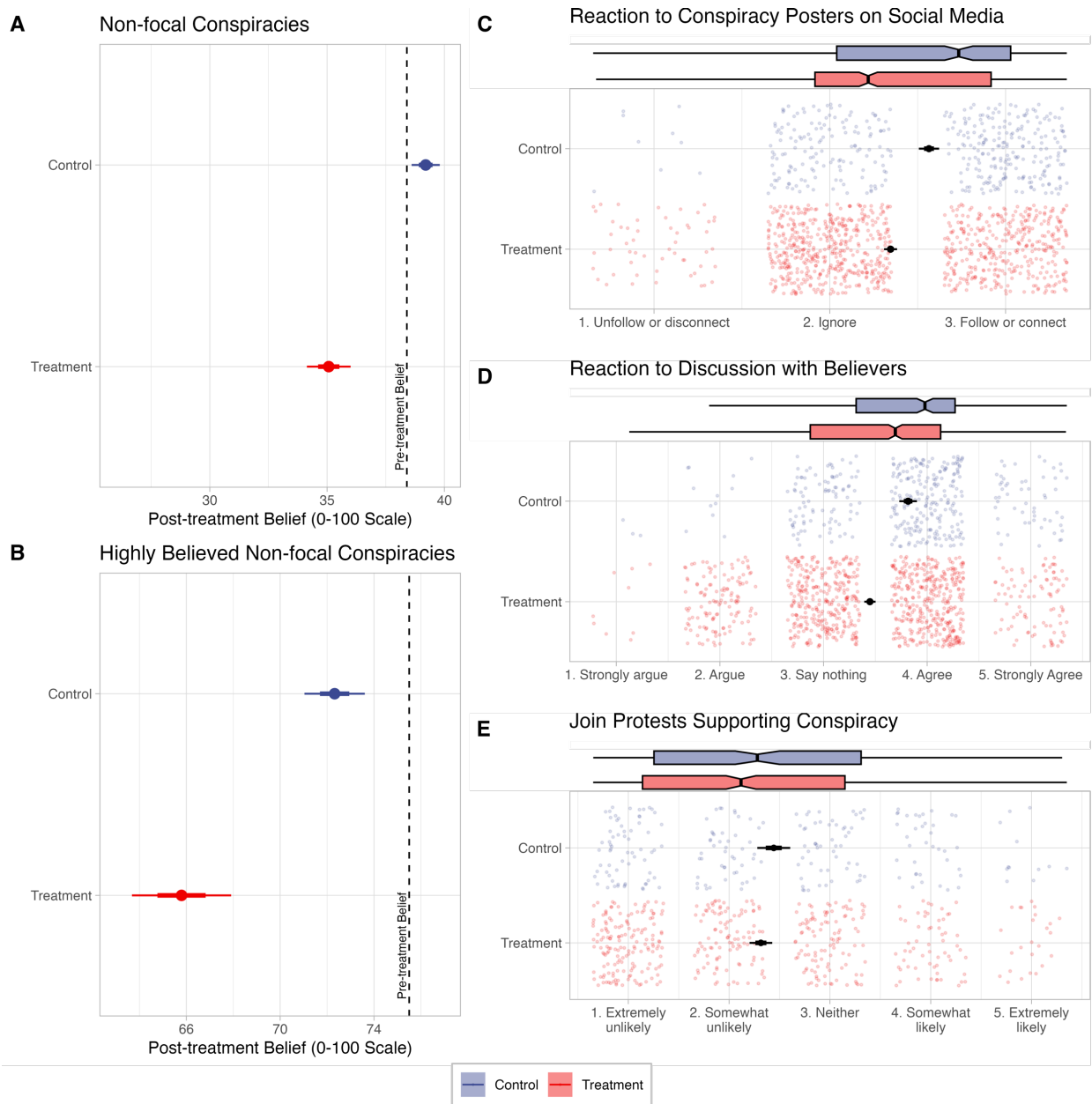


Figure 4. The treatment also affects belief in other conspiracy theories and behavioral intentions. First column: Post conversation average belief in the 15 conspiracies from the Belief in Conspiracy Theories Index (excluding the focal conspiracy, if it was one of those 15) by condition, for all conspiracies (A) and for only the subset of conspiracies with the participant indicated believing pre-treatment (B). Vertical dotted line indicates average pretreatment belief. Second column: Post-conversation behavioral intentions by condition. Shown are participants' intentions regarding how they would respond to social media users who espouse their focal conspiracy (C), how they would behave in conversation with someone who believes the focal conspiracy, and (D) how likely they would be to participate in a protest in support of the focal conspiracy. Thick error bars indicate 66% confidence intervals, thin error bars indicate 95% confidence intervals. Boxplots narrow at the median.

Discussion

While conspiracy theories are widely seen as a paradigmatic example of beliefs that rarely change in response to evidence (8–10), we hypothesized that dialogues with large language models – which can use facts and evidence to rebut the specific claims made by any given conspiracy believer – would be efficacious in debunking conspiracy beliefs. Our findings confirmed this prediction: a brief interaction with a pre-trained large language model substantially reduced belief in a wide range of conspiracy theories. The robustness of this effect is particularly noteworthy: (i) it occurred for both conspiracies that participants articulated in their own words and a broader conspiratorial worldview, (ii) it was evident even among participants with strong commitment to their chosen conspiracy, and (iii) its impact persisted, virtually undiminished, for (at least) two months after the intervention. Dialogues with the AI produced a meaningful and enduring shift in beliefs among a meaningful proportion of committed conspiracy believers in our study.

Theoretical, practical, and methodological advances

Our findings fundamentally challenge the view that evidence and arguments are of little use once someone has “gone down the rabbit hole” and come to believe a conspiracy theory. They also call into question social-psychological theories that center psychological “needs” and motivations as primary drivers of conspiratorial belief (1, 15, 42). Instead, our results align more closely with an alternative theoretical perspective that posits a central role for analytic thinking in protecting against epistemically suspect beliefs and behaviors (14), such as superstitions and paranormal beliefs (43), misinformation (44) and pseudo-profound bullshit (45). This viewpoint suggests that reasoning is not unduly constrained by identity needs and non-accuracy motivations; rather, people are generally willing to update their beliefs when presented with compelling evidence (46). Our study supports this perspective in several ways. Most straightforwardly, many conspiracists – including those strongly committed to their beliefs – updated their views when confronted with an AI that argued compellingly against their positions. Further, the AI primarily provided alternative, non-conspiratorial explanations and evidence while encouraging critical thinking, rather than attempting to satisfy psychological needs (see SM Section 6). The durability of our findings across two months, along with the intervention’s spillover effects on unrelated conspiracies and behavioral intentions, also suggests that participants seriously considered and internalized the AI’s arguments—consistent with the “central route” to persuasion (47), which is known to promote durable belief change (and in contrast to the “peripheral route” which leverages superficial identity cues or emotional appeals, and produces more ephemeral changes). Of course, our results do not wholly rule out some role for needs and motivations in the formation and maintenance of conspiracy beliefs, but they do indicate an important (perhaps countervailing, in some cases) role for evidence-based deliberation—especially in challenging and changing these beliefs once they are established. It is important to note that our goal – refuting existing conspiratorial beliefs – is distinct from other related challenges which have received more attention in the literature. These include presenting and then debunking pro-conspiracy arguments (48, 49), attempting to increase resistance to conspiracy theories in general (50, 51), and presenting arguments against a specific conspiracy theory to randomly selected crowd workers (52–55), which have been targeted with modest success by past work (e.g. meta-analytic $g = 0.16$ across 273 effect sizes) (12).

Our findings also have practical implications. Most broadly—in contrast to notions of a “post-truth” world in which facts no longer matter—arguments and evidence should not be abandoned by those seeking to reduce belief in dubious conspiracy theories. More specifically, AI models are powerful, flexible tools for reducing epistemically suspect beliefs and have the potential to be deployed to provide accurate information at scale. For example, internet search terms related to conspiracies could be met with an AI-generated summaries of accurate information—tailored to the precise search—that solicit the user’s response and engagement. Similarly, AI-powered social media accounts could reply to users who share inaccurate conspiracy-related content (providing corrective information for the potential benefit of both the poster and observers). Consistent with the potential for uptake of AI dialogues, some conspiracy-believing respondents in our sample expressed excitement and appreciation in their conversations with the AI (e.g., “Now this is the very first time I have gotten a response that made real, logical, sense. I must admit this really shifted my imagination when it comes to the subject of Illuminati. I think it was extremely helpful in my conclusion of rather the Illuminati is actually real.”). However, it is quite unlikely that all, or even many, entrenched believers will choose to engage with AI chatbots. Exploring a variety of short- and long-term strategies to encourage engagement—such as gamification, transparency efforts (e.g., disclosing the AI model prompt and fine-tuning; clearly labelling sources), incentive programs, anonymous interaction options, and the integration of AI-assisted critical thinking exercises into school curricula—is an important direction for future applied work.

The effectiveness of AI persuasion demonstrated in our studies also relates to ongoing debates regarding the promise versus peril of generative AI (56, 57). In our experiments, we sought to use AI to increase the accuracy of people’s beliefs by debunking conspiracy theories. Absent appropriate guardrails, however, it is entirely possible that such models could also convince people to adopt epistemically suspect beliefs (58) – or be used as tools of large-scale persuasion more generally (59). Thus, our findings emphasize both the potential positive impacts of generative AI when deployed responsibly, and the crucial, pressing importance of minimizing opportunities for this technology to be used irresponsibly. One especially key outstanding question, with far-reaching implications for AI’s impact on the global information ecosystem, is the degree of (a)symmetry in the efficacy of AI-based rational persuasion for true versus false content.

Finally, the experimental paradigm presented in this paper represents a substantial methodological advancement in behavioral science. Traditional survey experiments typically rely on static, predetermined stimuli and questions, which limits their ability to probe and respond to individuals’ beliefs (60). In contrast, the real-time use of LLMs embedded in a survey enables the researcher to elicit open-ended statements of belief (or anything else) and translate them into quantitative outcomes (61). As we have seen, AI can engage in back-and-forth dialogues with participants, adapting its responses based on the specific information provided by each individual (as opposed to, for example, using LLMs to pre-generate static stimuli as in past work) (62–66). This personalized approach is particularly valuable when studying complex phenomena such as conspiracy beliefs, where a one-size-fits-all intervention may be less effective (11, 12, 55, 67). The open-ended nature of the human-AI conversations also produces rich textual data that can be analyzed using natural language processing or qualitative techniques (68), which allows researchers to gain deeper insights into the content and structure of participants’ beliefs, as well as the strategies employed by the AI to challenge those beliefs.

Integrating human-LLM interactions into behavioral science has the potential to meaningfully enhance our understanding of complex psychological phenomena.

Limitations and future directions

Although our results are promising, there are important limitations to highlight. Our study primarily relied on American online survey respondents who chose to participate in studies for material compensation, which raises questions about generalizability. Future work should test whether our findings extend to conspiracy believers who do not typically participate in survey studies, as well as to populations from countries and cultures beyond the United States. While many participants in our study expressed maximal confidence in their conspiracy beliefs, it also remains to be seen whether AI dialogues would effectively change the beliefs of even more entrenched conspiracy adherents, such as those actively participating in conspiracy-related groups or events. Moreover, our use of GPT-4 Turbo, a frontier, closed-source, pre-trained, and fine-tuned language model, presents challenges related to interpretability and replicability (69–71). While GPT-4 demonstrated both high accuracy and persuasiveness, serving as a proof of concept for AI-driven debunking, it remains unknown whether other models would perform similarly along either or both dimensions (72). This uncertainty extends to the potentially interactive relationship between accuracy and persuasive capacity: hallucinations or lies may afford more compelling arguments, allowing models with less restrictive guardrails to outcompete heavily moderated models such as GPT-4 on persuasion. Finally, the causal mechanisms underpinning our results remain unformalized. While our study demonstrates the effectiveness of AI-facilitated dialogues in changing conspiracy beliefs, the specific cognitive or psychological processes through which this change occurs are unusually difficult to confirm—each conversation was unique and contained an admixture of rational argumentation and social cues. Both qualitative examination of the conversations and a structured, NLP-based analysis of the persuasive strategies used by the AI (see SM Section 6) suggest that fact-based argumentation was the focal point of each interaction; future research should examine this in greater detail.

Conclusion

It has become almost a truism that people “down the rabbit hole” of conspiracy belief are almost impossible to reach. In contrast to this pessimistic view, we have shown that a relatively brief conversation with a generative AI model can produce a large and lasting decrease in conspiracy beliefs, even among people whose beliefs are deeply entrenched. It may be that it has proven so difficult to dissuade people from their conspiracy beliefs because they simply have not been given good enough counterevidence. This paints a picture of human reasoning that is surprisingly optimistic: even the deepest of rabbit holes may have an exit. Conspiracists are not necessarily blinded by psychological needs and motivations – it just takes a genuinely strong argument to reach them.

Materials and Methods

All studies began by obtaining informed consent from participants. After the studies were completed, all participants were debriefed and informed about the limitations and constraints of generative AI models. All studies were deemed minimal risk and exempt by the MIT Committee on the Use of Humans as Experimental Subjects (protocol E-5539).

We excluded participants for inattentiveness (both before they entered the study, using an open-ended text response, and early on in the study before random assignment using an attention check item). All studies were preregistered (see aspredicted.org/RPG_RY9, aspredicted.org/HSD_41Q, aspredicted.org/KSN_PNL). Any non-pre-registered analyses are labeled “post hoc” and any deviations from the pre-registrations are reported. Conversational data from all participants, including those removed from our analyses, is available via [web application](#). All GPT-4 model prompts used during the experiment are provided in Table S2.

Study 1

Participants

We preregistered a target sample of 1,000 responses from CloudResearch’s Connect participant pool. In total, 1190 individuals began the survey (this includes 75 participants from a pilot conducted prior to the pre-registration; for completeness, we include these participants in our analyses, but excluding them does not qualitatively change the results). An initial (pre-treatment) screener only allowed participants who passed a permissive writing quality and coherence screener to continue and complete the survey. The purpose of this screening criterion was to ensure that participants were not using automated survey completion programs, were capable of reading and writing in English, and were willing to answer the sort of open-ended questions on which the intervention relies. Of the participants who entered the survey, 70 failed this writing screener. A further 13 participants failed pre-treatment attention checks and were removed from the survey; 86 discontinued prior to reaching the treatment. Further, using preregistered criteria, we excluded 157 participants who did not supply a genuine conspiracy theory (e.g., by noting that they do not believe any conspiracy theories in the open-ended response), 56 participants who provided a genuine conspiracy theory but endorsed it at below 50% veracity, and 55 participants for whom the AI provided an inaccurate summary (see SI section 1 and Figure S2). Thus, 774 participants were included in our analyses (although all who passed the writing screener were allowed to complete the experiment). The overall attrition rate was 1.8%. Using a logistic regression model predicting whether or not a person attrited, we find no evidence of differential rates of attrition in treatment vs. control ($b = -.53, p = .37$). The treatment sample (mean age = 45.7, mean ideology = 3.04 on a scale from 1 [liberal] to 6 [conservative]) included 383 males, 384 females, and 7 participants who selected another gender option. A balance check found that our sample was balanced on pre-treatment covariates (see Table S1). This study was run on 19-22 January 2024 and took 30.98 minutes on average to complete.

Procedure

Pre-treatment measures. Participants completed a battery of self-report measures concerning their endorsement of a diverse set of 15 conspiracy beliefs, their attitudes concerning artificial

intelligence, and demographic items including beliefs about politics and religion. Conspiracy beliefs were assessed using a modified version of Belief in Conspiracy Theories Inventory ($\alpha = .90$; example item: “Government agencies in the UK are involved in the distribution of illegal drugs to ethnic minorities”) (73), which updated several items to reflect contemporary versions of the original (e.g., “SARS” was swapped with “COVID-19”). The scale labels ranged from “0 (Definitely False)” to “25 (Probably False)” to “50 (Uncertain)” to “75 (Probably True)” to “100 (Definitely True)”, with the mean score in the treatment sample being 38.6% (SD = 20.0%). In addition to the 15 false conspiracy theories comprising the BCTI, we included 3 true conspiracy theories (pertaining to Project MK Ultra, Operation Northwoods, and the tobacco industry). Attitudes concerning artificial intelligence were measured using items adapted from a Pew survey (74).

Subsequently, participants responded to an open-ended question concerning a conspiracy theory that they support (which we refer to as the “focal conspiracy”):

“What is a significant conspiracy theory that you find credible and compelling? Could you please describe this theory and share why it resonates with you?”

They then were asked to elaborate on the next page:

“On the previous question, you wrote [RESPONSE]. Can you describe in detail the specific evidence or events that initially led you to believe in this conspiracy theory? How do you interpret this evidence in relation to commonly accepted explanations for the same events?”

This information was fed forward to an instance of GPT-4 Turbo, which was tasked with summarizing the conspiratorial belief into a single sentence (see Table S4 for the exact wording of this API query). Participants were then asked to rate their belief in the summarized conspiracy’s veracity (“Please indicate your level of confidence that this statement is true”) using a scale that ranged from 0 (“Definitely False”) to 25 (“Probably False”) to 50 (“Uncertain”) to 75 (“Probably True”) to 100 (“Definitely True”).

Human-AI dialogues. Following these pre-treatment measures, participants were informed they would be conversing with an advanced AI. To facilitate this real-time interaction within the Qualtrics survey platform, we used JavaScript to call OpenAI’s Chat Completions API, dynamically inject participant-specific information into the model’s instructions, and display the model’s responses. Several details of this approach are worth mentioning. First, we used the latest available GPT-4 model, which was gpt-4-1106-preview for Study 1 and gpt-4-0125-preview for Study 2. Second, conversations were formatted to begin with a system message, followed by alternating user and AI messages. The system message, in our case, included details about the context, the goal of refuting a conspiracy belief, and instructions for how the model should behave (all of which were invariant across each model call), as well as the participant’s specific conspiracy theory, the participant’s stated reasons for believing that theory, and the participant’s level of belief in the conspiracy (which varied across participants). Otherwise, our model instructions were simple and did not involve hidden reasoning steps (e.g., chain-of-thought) or access to external tools (e.g., internet browsing) beyond those provided by default. To facilitate a continuous conversation, for rounds 2 and 3 the previous AI- and human-

messages were included in the prompt as conversation histories. Third, no token limit was placed on the AI's responses, which frequently comprised hundreds of words (Figure S3), multiple paragraphs, and markdown formatting (e.g., lists and section headings). Thus, although each dialogue only lasted 3 rounds, the dialogues (a) represented 8.4 minutes of AI-human engagement on average and (b) were information-dense, yet comparatively easy to read and parse. Fourth, the AI's messages were sent to participants after the full response was constructed (rather than streamed word by word), necessitating idle time between each round of dialogue during which a loading screen was shown.

In the treatment condition, the AI was instructed to argue persuasively against the participant's conspiracy theory. In the control conditions, the AI was instructed to either (a) discuss the American medical system, (b) debate with participants about whether they prefer dogs or cats, or (c) discuss participants' past experiences with firefighters. We used a 60/40 split when randomizing participants into the treatment or control conditions, and participants assigned to the control were further randomized to one of the three control conditions, such that roughly 13-14% of the sample was assigned to each control condition. No significant differences were identified across the control groups, so we pooled them for all subsequent analyses.

Post-treatment measures. Following the conversations, participants re-rated their belief in the focal conspiracy and then again completed the modified BCTI ($\alpha = .92$). Given that, in many cases, participants' focal conspiracies resembled at least one item on the BCTI (the items were chosen to reflect the most popular conspiracy theories), we computed three versions of pre- and post-treatment BCTI scores. The first version was the mean response on all 15 BCTI items, which we used to identify participants with a highly conspiratorial worldview. In the second version, we dropped items that matched the participants' focal conspiracy theory. Overlap was identified using an instance of GPT-4 that was supplied with each participant's conspiracy and each BCTI item and queried concerning which of the BCTI items reflected an affirmative belief in the participant's conspiracy using a binary judgement (see SI section 7), yielding overlap-adjusted BCTI scores for pre-treatment ($\alpha = .90$) and post-treatment ($\alpha = .92$). Thirdly, we further filtered the BCTI item pool by retaining non-overlapping items that participants initially rated above 50% (more belief than "uncertain"), which allowed for pre-treatment ($\alpha = .90$) and post-treatment ($\alpha = .90$) overlap-adjusted BCTI scores for conspiracy theories that each participant actively endorsed. We also administered the three true conspiracy items.

Recontacting at 10-days and 2-months. The participants from Study 1 were recontacted twice. The first recontact occurred 10 days after completing the intervention (T3; $n = 631$, dropout rate = 15.7% and 15.6% for the treatment and control groups, respectively). Participants in the treatment condition who completed the T3 follow-up did not significantly differ from those who did not return for either pre-treatment belief in their chosen conspiracy ($t[454] = 0.61, p = .544$) or on the pre-treatment BCTI ($t[454] = -0.71, p = .475$). Participants completed the same dependent variables as in Study 1 (i.e., endorsement of their chosen conspiracy theory and the BCTI). The second recontact occurred 2 months (T4) after completing the intervention ($n = 529$, dropout rate = 32.1% and 31.1% for the treatment and control groups, respectively). As with T3, participants in the treatment who remained did not differ from those who dropped for either pre-treatment belief in their chosen conspiracy ($t[450] = 0.02, p = .977$) or on the pre-treatment BCTI ($t[450] = -1.33, p = .183$).

Study 2

For Study 2, two additional samples (Study 2a and 2b) were fielded from CloudConnect to corroborate, replicate, and extend our experimental findings. Although the majority of materials were identical across Studies 2a and 2b, we describe them separately because (a) we pre-registered separate rounds of data collection, (b) we used different phrasings for the behavioral outcome items, and (c) the data were collected several weeks apart. Particularly, we collected Study 2b due to imprecise wording used in certain behavioral outcome items in Study 2a, as noted below. In the main text, results are pooled across Studies 2a and 2b, except for those pertaining to the behavioral outcomes that were modified between 2a and 2b.

Participants

In Study 2a, we preregistered a target sample of 1,000 complete responses from CloudResearch's Connect participant pool, using quota-based sampling for age, race, ethnicity, and gender. A total of 1,427 individuals entered the survey, of whom 312 were redirected for using a cell phone, 30 failed the initial pre-treatment writing screener, 14 failed an attention check, and 104 discontinued prior to treatment, leaving 968. Of these participants, 218 did not provide a genuine conspiracy theory and 81 endorsed their conspiracy statement at below 50% certainty – such that the final sample analyzed sample size was $n = 668$. Similarly, in Study 2b, we recruited 1545 demographically representative participants using the Connect pool, of whom 27 were redirected for using a cell phone, 30 failed the writing screen, 27 failed an attention check, and 152 discontinued prior to treatment, leaving 1309. Of these participants, 296 did not provide a genuine conspiracy theory and another 128 did not endorse their conspiracy above 50%, leaving a treatment sample of 885.

Thus, the full sample size across both rounds of Study 2 was $n = 1553$ (mean age = 41.9, mean ideology = 3.09), which included 670 males, 724 females, and 13 participants who selected another gender option (see Figure S4 and Table S3). These studies were run on 25-28 February and 4-9 March 2024 and took 24.4 and 27.85 minutes on average to complete. The overall attrition rate was 3.7%. Using a logistic regression model predicting whether or not a person attrited, we find no evidence of differential rates of attrition in treatment vs. control ($b = .02, p = .97$).

Procedure

Pre-treatment measures. For all open-ended responses, including those in the Human-AI dialogues, the “paste” functionality was disabled to prevent automated responding. As in Study 1, participants began the experiment by answering a simple, writing-intensive question designed to gauge their willingness and ability to take part in a written conversation. Those whose responses were determined by GPT-4 Turbo to be low-effort or incoherent were redirected from the survey. Subsequently, participants completed self-report items about their artificial intelligence attitudes and demographic characteristics (mirroring those from Study 1). We did not administer the Belief in Conspiracy Theories Index in Study 2, and instead proceeded directly to the person-specific conspiracy assessment.

The wording of the person-specific instructions were modified slightly from Study 1 to (a) explicitly define the theories to be described and (b) only indirectly classify the theories as “conspiracies”. The first question’s wording was:

“Throughout history, various theories have emerged that suggest certain significant events or situations are the result of secret plans by individuals or groups. These theories often offer alternative explanations for events than those that are widely accepted by the public or presented by official sources. Some people call these ‘conspiracy theories’. Reflecting on this, are there any specific such theories that you find particularly credible or compelling? Please describe one below and share your reasons for finding it compelling.”

And the follow-up question, presented on a separate page:

“On the previous question, you wrote: “[conspiracy]”. Could you share more about what led you to find this theory compelling? For instance, are there specific pieces of evidence, events, sources of information, or personal experiences that have particularly influenced your perspective? Please describe these in as much detail as you feel comfortable.”

As in Study 1, this information was fed forward to an instance of GPT-4 Turbo, which was tasked with summarizing the conspiratorial belief into a single sentence. Participants then provided a rating reflecting their confidence in the summarized statement’s truth. The vast majority (90.6%) reported that the AI model accurately summarized their perspective; participants who received inaccurate summaries were excluded from subsequent analysis (note that this is a pre-treatment exclusion). Before proceeding to the treatment, participants reported how important the conspiracy was to them (“How important is this theory to your personal beliefs or understanding of the world?”) on a scale from 0 (“Not all all important to my beliefs and worldview”) to 8 (“Extremely important to my beliefs and worldview”).

Post-treatment measures. Following the conversations, participants re-rated the focal conspiracy’s veracity and then completed a set of measures related to conspiracy-relevant behavior and trust. In both studies, we assessed (a) intentions to ignore or unfollow social media accounts espousing the focal conspiracy and (b) willingness to ignore or argue against people who believe the focal conspiracy; in our analyses of these items, we pool data across Studies 2a and 2b. Study 2a also asked about (c) willingness to engage in collective actions opposing the focal conspiracy, and (d) intentions to join protests related to the focal conspiracy theory. After data collection, however, we noticed problems in the wording of these items that made them uninterpretable, and thus we do not analyze these items. Item c, concerning collective actions, was both counter-directionally worded (relative to the other items) and used a response scale containing negative and positive options that was *not* counter-directionally worded, potentially resulting in a confused pattern of results. Item d, reflecting protest intentions, did not specify whether the protests supported or opposed the focal conspiracy, making responses to that item uninterpretable. In Study 2b, we attempted to rectify these issues by dropping item c and changing the wording of item d to remove the ambiguity (i.e., “If people you knew were going to engage in a protest or action in support of the theory you described, how likely would you be to join in?”), as well as visually highlighting words indicating item directionality and having response-option direction randomized between participants and standardized within participants.

Finally, in Study 2b we asked GPT-4 Turbo to generate petitions *opposing* the participants’ focal conspiracy theory, which we then asked participants if they wanted to sign. Unfortunately,

inspecting these petitions indicated that many of them were not actually in opposition to the focal conspiracy theory (e.g. for a participant who thought the government was concealing the existence of aliens, GPT-4 Turbo asked if they wanted to sign a petition calling for greater government transparency about aliens – which plays into the conspiracy theory, rather than opposing it). To determine how serious of a problem this was, we conducted a post hoc analysis in which 670 crowd workers each rated a random subset of 3 petitions as either “opposing” or “not opposing” its corresponding conspiracy theory after completing a brief training exercise. Of the 404 petitions rated at least twice, only 199 (49.3%) were rated as actually opposing the focal conspiracy in more than half of responses; and only 118 (29.2%) were unanimously rated as opposing the conspiracy. This makes participants’ choice of whether to sign the petition not useful for determining the effect of the intervention, and thus we do not include analysis of it.

In both Study 2a and 2b, participants then completed measures of general trust (1-item), personal trust (1-item), and institutional trust (5-items), which were adapted from the OECD Guidelines on Measuring Trust (75).

References and notes

1. S. M. Bowes, T. H. Costello, A. Tasimi, The conspiratorial mind: A meta-analytic review of motivational and personological correlates. *Psychol. Bull.* (2023).
2. M. Butter, P. Knight, Eds., *Routledge Handbook of Conspiracy Theories* (Routledge, London, 2020).
3. K. M. Douglas, R. M. Sutton, What Are Conspiracy Theories? A Definitional Approach to Their Correlates, Consequences, and Communication. *Annu. Rev. Psychol.* **74**, 271–298 (2023).
4. J. E. Oliver, T. J. Wood, Conspiracy Theories and the Paranoid Style(s) of Mass Opinion. *Am. J. Polit. Sci.* **58**, 952–966 (2014).
5. H. G. West, T. Sanders, *Transparency and Conspiracy: Ethnographies of Suspicion in the New World Order* (Duke University Press, 2003).
6. J. E. Uscinski, J. M. Parent, *American Conspiracy Theories* (Oxford University Press, 2014).
7. J.-W. van Prooijen, K. M. Douglas, Belief in conspiracy theories: Basic principles of an emerging research domain. *Eur. J. Soc. Psychol.* **48**, 897–908 (2018).
8. S. Lewandowsky, G. E. Gignac, K. Oberauer, The Role of Conspiracist Ideation and Worldviews in Predicting Rejection of Science. *PLoS ONE* **8**, e75637 (2013).
9. M. G. Napolitano, “Conspiracy Theories and Resistance to Evidence,” thesis, UC Irvine (2022).
10. C. R. Sunstein, A. Vermeule, Conspiracy Theories: Causes and Cures. *J. Polit. Philos.* **17**, 202–227 (2008).
11. C. O’Mahony, M. Brassil, G. Murphy, C. Linehan, The efficacy of interventions in reducing belief in conspiracy theories: A systematic review. *PLoS ONE* **18**, e0280902 (2023).
12. L. Stasielowicz, *How to Reduce Conspiracy Beliefs? A Meta-Analysis of Intervention Studies* (2024).
13. J. K. Madsen, L. de-Wit, P. Ayton, C. Brick, L. de-Moliere, C. J. Groom, Behavioral science should start by assuming people are reasonable. *Trends Cogn. Sci.*, doi: 10.1016/j.tics.2024.04.010 (2024).
14. G. Pennycook, “Chapter Three - A framework for understanding reasoning errors: From fake news to climate change and beyond” in *Advances in Experimental Social Psychology*, B. Gawronski, Ed. (Academic Press, 2023; <https://www.sciencedirect.com/science/article/pii/S0065260122000284>)vol. 67, pp. 131–208.
15. K. M. Douglas, R. M. Sutton, A. Cichocka, The Psychology of Conspiracy Theories. *Curr. Dir. Psychol. Sci.* **26**, 538–542 (2017).
16. J. T. Jost, A. Ledgerwood, C. D. Hardin, Shared Reality, System Justification, and the Relational Basis of Ideological Beliefs. *Soc. Personal. Psychol. Compass* **2**, 171–186 (2008).
17. R. Hofstadter, *The Paranoid Style in American Politics* (Knopf Doubleday Publishing Group, 1964).
18. J. A. Whitson, A. D. Galinsky, Lacking Control Increases Illusory Pattern Perception. *Science* **322**, 115–117 (2008).
19. S. Lewandowsky, J. Cook, K. Oberauer, S. Brophy, E. A. Lloyd, M. Marriott, Recurrent fury: Conspiratorial discourse in the blogosphere triggered by research on the role of conspiracist ideation in climate denial. *J. Soc. Polit. Psychol.* **3**, 142–178 (2015).
20. J.-W. van Prooijen, N. B. Jostmann, Belief in conspiracy theories: The influence of uncertainty and perceived morality. *Eur. J. Soc. Psychol.* **43**, 109–115 (2013).
21. J.-W. van Prooijen, An Existential Threat Model of Conspiracy Theories. *Eur. Psychol.* **25**, 16–25 (2020).
22. A. Lantian, D. Muller, C. Nurra, K. M. Douglas, “I Know Things They Don’t Know!” *Soc. Psychol.* **48**, 160–173 (2017).
23. M. Biddlestone, R. Green, A. Cichocka, K. Douglas, F. Azevedo, R. Sutton, Reasons to believe: A systematic review and meta-analytic synthesis of the motives associated with conspiracy beliefs. OSF [Preprint] (2022). <https://doi.org/10.31234/osf.io/rxjqc>.
24. M. Biddlestone, R. Green, A. Cichocka, R. Sutton, K. Douglas, Conspiracy beliefs and the individual, relational, and collective selves. *Soc. Personal. Psychol. Compass* **15**, e12639 (2021).
25. A. Cichocka, M. Marchlewska, A. Golec de Zavala, M. Olechowski, ‘They will not control us’: Ingroup positivity and belief in intergroup conspiracies. *Br. J. Psychol.* **107**, 556–576 (2016).
26. A. Sternisko, A. Cichocka, A. Cislak, J. J. Van Bavel, National Narcissism predicts the Belief in and the Dissemination of Conspiracy Theories During the COVID-19 Pandemic: Evidence From 56 Countries. *Pers. Soc. Psychol. Bull.* **49**, 48–65 (2023).
27. R. Brotherton, *Suspicious Minds: Why We Believe Conspiracy Theories* (Bloomsbury Publishing, 2015).
28. R. K. Garrett, B. E. Weeks, Epistemic beliefs’ role in promoting misperceptions and conspiracist ideation. *PLoS ONE* **12**, e0184733 (2017).

29. N. Dagnall, K. Drinkwater, A. Parker, A. Denovan, M. Parton, Conspiracy theory and cognitive style: a worldview. *Front. Psychol.* **6** (2015).
30. S. Novella, *The Skeptics' Guide to the Universe: How to Know What's Really Real in a World Increasingly Full of Fake* (Hachette UK, 2018).
31. P. M. Fernbach, J. E. Bogard, Conspiracy Theory as Individual and Group Behavior: Observations from the Flat Earth International Conference. *Top. Cogn. Sci.* **n/a**.
32. H. Wang, J. Li, H. Wu, E. Hovy, Y. Sun, Pre-Trained Language Models and Their Applications. *Engineering* **25**, 51–65 (2023).
33. OpenAI, J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, R. Avila, I. Babuschkin, S. Balaji, V. Balcom, P. Baltescu, H. Bao, M. Bavarian, J. Belgum, I. Bello, J. Berdine, G. Bernadett-Shapiro, C. Berner, L. Bogdonoff, O. Boiko, M. Boyd, A.-L. Brakman, G. Brockman, T. Brooks, M. Brundage, K. Button, T. Cai, R. Campbell, A. Cann, B. Carey, C. Carlson, R. Carmichael, B. Chan, C. Chang, F. Chantzis, D. Chen, S. Chen, R. Chen, J. Chen, M. Chen, B. Chess, C. Cho, C. Chu, H. W. Chung, D. Cummings, J. Currier, Y. Dai, C. Decareaux, T. Degry, N. Deutsch, D. Deville, A. Dhar, D. Dohan, S. Dowling, S. Dunning, A. Ecoffet, A. Eleti, T. Eloundou, D. Farhi, L. Fedus, N. Felix, S. P. Fishman, J. Forte, I. Fulford, L. Gao, E. Georges, C. Gibson, V. Goel, T. Gogineni, G. Goh, R. Gontijo-Lopes, J. Gordon, M. Grafstein, S. Gray, R. Greene, J. Gross, S. S. Gu, Y. Guo, C. Hallacy, J. Han, J. Harris, Y. He, M. Heaton, J. Heidecke, C. Hesse, A. Hickey, W. Hickey, P. Hoeschele, B. Houghton, K. Hsu, S. Hu, X. Hu, J. Huizinga, S. Jain, S. Jain, J. Jang, A. Jiang, R. Jiang, H. Jin, D. Jin, S. Jomoto, B. Jonn, H. Jun, T. Kaftan, L. Kaiser, A. Kamali, I. Kanitscheider, N. S. Keskar, T. Khan, L. Kilpatrick, J. W. Kim, C. Kim, Y. Kim, J. H. Kirchner, J. Kiros, M. Knight, D. Kokotajlo, L. Kondraciuk, A. Kondrich, A. Konstantinidis, K. Kosic, G. Krueger, V. Kuo, M. Lampe, I. Lan, T. Lee, J. Leike, J. Leung, D. Levy, C. M. Li, R. Lim, M. Lin, S. Lin, M. Litwin, T. Lopez, R. Lowe, P. Lue, A. Makanju, K. Malfacini, S. Manning, T. Markov, Y. Markovski, B. Martin, K. Mayer, A. Mayne, B. McGrew, S. M. McKinney, C. McLeavey, P. McMillan, J. McNeil, D. Medina, A. Mehta, J. Menick, L. Metz, A. Mishchenko, P. Mishkin, V. Monaco, E. Morikawa, D. Mossing, T. Mu, M. Murati, O. Murk, D. Mély, A. Nair, R. Nakano, R. Nayak, A. Neelakantan, R. Ngo, H. Noh, L. Ouyang, C. O'Keefe, J. Pachocki, A. Paino, J. Palermo, A. Pantuliano, G. Parascandolo, J. Parish, E. Parparita, A. Passos, M. Pavlov, A. Peng, A. Perelman, F. de A. B. Peres, M. Petrov, H. P. de O. Pinto, Michael, Pokorny, M. Pokrass, V. H. Pong, T. Powell, A. Power, B. Power, E. Proehl, R. Puri, A. Radford, J. Rae, A. Ramesh, C. Raymond, F. Real, K. Rimbach, C. Ross, B. Rotsted, H. Roussez, N. Ryder, M. Saltarelli, T. Sanders, S. Santurkar, G. Sastry, H. Schmidt, D. Schnurr, J. Schulman, D. Selsam, K. Sheppard, T. Sherbakov, J. Shieh, S. Shoker, P. Shyam, S. Sidor, E. Sigler, M. Simens, J. Sitkin, K. Slama, I. Sohl, B. Sokolowsky, Y. Song, N. Staudacher, F. P. Such, N. Summers, I. Sutskever, J. Tang, N. Tezak, M. B. Thompson, P. Tillet, A. Tootoonchian, E. Tseng, P. Tuggle, N. Turley, J. Tworek, J. F. C. Uribe, A. Vallone, A. Vijayvergiya, C. Voss, C. Wainwright, J. J. Wang, A. Wang, B. Wang, J. Ward, J. Wei, C. J. Weinmann, A. Welihinda, P. Welinder, J. Weng, L. Weng, M. Wiethoff, D. Willner, C. Winter, S. Wolrich, H. Wong, L. Workman, S. Wu, J. Wu, M. Wu, K. Xiao, T. Xu, S. Yoo, K. Yu, Q. Yuan, W. Zaremba, R. Zellers, C. Zhang, M. Zhang, S. Zhao, T. Zheng, J. Zhuang, W. Zhuk, B. Zoph, GPT-4 Technical Report. arXiv arXiv:2303.08774 [Preprint] (2024). <https://doi.org/10.48550/arXiv.2303.08774>.
34. K. Arceneaux, B. N. Bakker, N. Fasching, Y. Lelkes, A critical evaluation and research agenda for the study of psychological dispositions and political attitudes. *Polit. Psychol.* **n/a**.
35. W. Yaqub, O. Kakhidze, M. L. Brockman, N. Memon, S. Patil, "Effects of Credibility Indicators on Social Media News Sharing Intent" in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (ACM, Honolulu HI USA, 2020; <https://dl.acm.org/doi/10.1145/3313831.3376213>), pp. 1–14.
36. R. Böhm, M. Jörling, L. Reiter, C. Fuchs, People devalue generative AI's competence but not its advice in addressing societal and personal challenges. *Commun. Psychol.* **1**, 1–10 (2023).
37. Y. Zhang, R. Gosline, Human favoritism, not AI aversion: People's perceptions (and bias) toward generative AI, human experts, and human–GAI collaboration in persuasive content generation. *Judgm. Decis. Mak.* **18**, e41 (2023).
38. V. Veselovsky, M. H. Ribeiro, P. Cozzolino, A. Gordon, D. Rothschild, R. West, Prevalence and prevention of large language model use in crowd work. arXiv arXiv:2310.15683 [Preprint] (2023). <https://doi.org/10.48550/arXiv.2310.15683>.
39. M. N. Stagnaro, D. G. Rand, "The Coevolution of Religious Belief and Intuitive Cognitive Style via Individual-Level Selection" in *The Oxford Handbook of Evolutionary Psychology and Religion*, J. R. Liddle, T. K. Shackelford, Eds. (Oxford University Press, 2016; <https://doi.org/10.1093/oxfordhb/9780199397747.013.10>), pp. 153–173.
40. S. Athey, J. Tibshirani, S. Wager, Generalized random forests. *Ann. Stat.* **47**, 1148–1178 (2019).

41. S. Hughes, M. Bae, M. Li, Vectara Hallucination Leaderboard, (2023); <https://github.com/vectara/hallucination-leaderboard>.
42. M. J. Hornsey, K. Bierwiazzonek, K. Sassenberg, K. M. Douglas, Individual, intergroup and nation-level influences on belief in conspiracy theories. *Nat. Rev. Psychol.* **2**, 85–97 (2023).
43. G. Pennycook, J. A. Cheyne, P. Seli, D. J. Koehler, J. A. Fugelsang, Analytic cognitive style predicts religious and paranormal belief. *Cognition* **123**, 335–346 (2012).
44. G. Pennycook, D. G. Rand, Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition* **188**, 39–50 (2019).
45. G. Pennycook, J. A. Cheyne, N. Barr, D. J. Koehler, J. A. Fugelsang, On the reception and detection of pseudo-profound bullshit. *Judgm. Decis. Mak.* **10**, 549–563 (2015).
46. B. M. Tappin, A. J. Berinsky, D. G. Rand, Partisans' receptivity to persuasive messaging is undiminished by countervailing party leader cues. *Nat. Hum. Behav.* **7**, 568–582 (2023).
47. R. E. Petty, J. T. Cacioppo, "The Elaboration Likelihood Model of Persuasion" in *Communication and Persuasion: Central and Peripheral Routes to Attitude Change*, R. E. Petty, J. T. Cacioppo, Eds. (Springer, New York, NY, 1986; https://doi.org/10.1007/978-1-4612-4964-1_1), pp. 1–24.
48. E. Porter, Y. Velez, T. J. Wood, Factual Corrections Eliminate False Beliefs About COVID-19 Vaccines. *Public Opin. Q.* **86**, 762–773 (2022).
49. G. Orosz, P. Krekó, B. Paskuj, I. Tóth-Király, B. Bóthe, C. Roland-Lévy, Changing Conspiracy Beliefs through Rationality and Ridiculing. *Front. Psychol.* **7** (2016).
50. J. A. Banas, G. Miller, Inducing Resistance to Conspiracy Theory Propaganda: Testing Inoculation and Metainoculation Strategies. *Hum. Commun. Res.* **39**, 184–207 (2013).
51. E. Bonetto, J. Troïan, F. Varet, G. Lo Monaco, F. Girandola, Priming Resistance to Persuasion decreases adherence to Conspiracy Theories*. *Soc. Infl.* **13**, 125–136 (2018).
52. V. Swami, J. Pietschnig, U. S. Tran, I. W. Nader, S. Stieger, M. Voracek, Lunar Lies: The Impact of Informational Framing and Individual Differences in Shaping Conspiracist Beliefs About the Moon Landings. *Appl. Cogn. Psychol.* **27**, 71–80 (2013).
53. D. Jolley, K. M. Douglas, Prevention is better than cure: Addressing anti-vaccine conspiracy theories. *J. Appl. Soc. Psychol.* **47**, 459–469 (2017).
54. S. Altay, A.-S. Hacquin, C. Chevallier, H. Mercier, Information delivered by a chatbot has a positive impact on COVID-19 vaccines attitudes and intentions. *J. Exp. Psychol. Appl.* **29**, 52–62 (2023).
55. S. Altay, M. Schwartz, A.-S. Hacquin, A. Allard, S. Blancke, H. Mercier, Scaling up interactive argumentation by providing counterarguments with a chatbot. *Nat. Hum. Behav.* **6**, 579–592 (2022).
56. E. Klein, Opinion | This Changes Everything, *The New York Times* (2023). <https://www.nytimes.com/2023/03/12/opinion/chatbots-artificial-intelligence-future-weirdness.html>.
57. D. Allen, E. G. Weyl, The Real Dangers of Generative AI. *J. Democr.* **35**, 147–162 (2024).
58. M. Phuong, M. Aitchison, E. Catt, S. Cogan, A. Kaskasoli, V. Krakovna, D. Lindner, M. Rahtz, Y. Assael, S. Hodgkinson, H. Howard, T. Lieberum, R. Kumar, M. A. Raad, A. Webson, L. Ho, S. Lin, S. Farquhar, M. Hutter, G. Deletang, A. Ruoss, S. El-Sayed, S. Brown, A. Dragan, R. Shah, A. Dafoe, T. Shevlane, Evaluating Frontier Models for Dangerous Capabilities. arXiv arXiv:2403.13793 [Preprint] (2024). <https://doi.org/10.48550/arXiv.2403.13793>.
59. M. Burtell, T. Woodside, Artificial Influence: An Analysis Of AI-Driven Persuasion. arXiv arXiv:2303.08721 [Preprint] (2023). <https://doi.org/10.48550/arXiv.2303.08721>.
60. Y. Velez, Crowdsourced Adaptive Surveys. arXiv arXiv:2401.12986 [Preprint] (2024). <https://doi.org/10.48550/arXiv.2401.12986>.
61. Y. R. Velez, P. Liu, Confronting Core Issues: A Critical Test of Attitude Polarization. *Am. Polit. Sci. Rev.* (2024).
62. H. Bai, J. Voelkel, J. Eichstaedt, R. Willer, Artificial intelligence can persuade humans on political issues. (2023).
63. E. Karinshak, S. X. Liu, J. S. Park, J. T. Hancock, Working With AI to Persuade: Examining a Large Language Model's Ability to Generate Pro-Vaccination Messages. *Proc. ACM Hum.-Comput. Interact.* **7**, 116:1–116:29 (2023).
64. K. Hackenburg, H. Margetts, Evaluating the persuasive influence of political microtargeting with large language models. (2023).
65. S. C. Matz, J. D. Teeny, S. S. Vaid, H. Peters, G. M. Harari, M. Cerf, The potential of generative AI for personalized persuasion at scale. *Sci. Rep.* **14**, 4692 (2024).

66. E. Durmus, L. Lovitt, A. Tamkin, S. Ritchie, J. Clark, D. Ganguli, Measuring the Persuasiveness of Language Models (2024). <https://www.anthropic.com/news/measuring-model-persuasiveness>.
67. M. N. Williams, M. Ling, J. R. Kerr, S. R. Hill, M. D. Marques, H. Mawson, E. J. R. Clarke, People do change their beliefs about conspiracy theories—but not often. *Sci. Rep.* **14**, 3836 (2024).
68. C. Olah, A. Jermyn, Reflections on Qualitative Research, *Transformer Circuits*. <https://transformer-circuits.pub/2024/qualitative-essay/index.html>.
69. Z. Hussain, M. Binz, R. Mata, D. U. Wulff, A tutorial on open-source large language models for behavioral science. OSF [Preprint] (2023). <https://doi.org/10.31234/osf.io/f7stn>.
70. A. Spirling, Why open-source generative AI models are an ethical way forward for science. *Nature* **616**, 413–413 (2023).
71. I. Grossmann, M. Feinberg, D. C. Parker, N. A. Christakis, P. E. Tetlock, W. A. Cunningham, AI and the transformation of social science research. *Science* **380**, 1108–1109 (2023).
72. K. Hackenburg, B. M. Tappin, P. Röttger, S. Hale, J. Bright, H. Margetts, Evidence of a log scaling law for political persuasion with large language models. arXiv arXiv:2406.14508 [Preprint] (2024). <https://doi.org/10.48550/arXiv.2406.14508>.

Supplementary Materials References

73. V. Swami, T. Chamorro-Premuzic, A. Furnham, Unanswered questions: A preliminary investigation of personality and individual difference predictors of 9/11 conspiracist beliefs. *Appl. Cogn. Psychol.* **24**, 749–761 (2010).
74. M. Faverio, A. Tyson, What the data says about Americans’ views of artificial intelligence, *Pew Research Center*. <https://www.pewresearch.org/short-reads/2023/11/21/what-the-data-says-about-americans-views-of-artificial-intelligence/>.
75. OECD, *OECD Guidelines on Measuring Trust* (Organisation for Economic Co-operation and Development, Paris, 2017; https://www.oecd-ilibrary.org/governance/oecd-guidelines-on-measuring-trust_9789264278219-en).
76. W. Lin, Agnostic notes on regression adjustments to experimental data: Reexamining Freedman’s critique. *Ann. Appl. Stat.* **7**, 295–318 (2013).
77. K. D. Carlson, F. L. Schmidt, Impact of experimental design on effect size: Findings from the research literature on training. *J. Appl. Psychol.* **84**, 851–862 (1999).
78. M. C. Fox, K. A. Ericsson, R. Best, Do procedures for verbal reporting of thinking have to be reactive? A meta-analysis and recommendations for best reporting methods. *Psychol. Bull.* **137**, 316–344 (2011).
79. C. Roberts, E. Gilbert, N. Allum, L. Eisner, Research Synthesis: Satisficing in Surveys: A Systematic Review of the Literature. *Public Opin. Q.* **83**, 598–626 (2019).
80. A. Neelakantan, T. Xu, R. Puri, A. Radford, J. M. Han, J. Tworek, Q. Yuan, N. Tezak, J. W. Kim, C. Hallacy, J. Heidecke, P. Shyam, B. Power, T. E. Niekoul, G. Sastry, G. Krueger, D. Schnurr, F. P. Such, K. Hsu, M. Thompson, T. Khan, T. Sherbakov, J. Jang, P. Welinder, L. Weng, Text and Code Embeddings by Contrastive Pre-Training. arXiv arXiv:2201.10005 [Preprint] (2022). <https://doi.org/10.48550/arXiv.2201.10005>.
81. M. Ester, H.-P. Kriegel, X. Xu, A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise.
82. M. Hahsler, M. Piekenbrock, D. Doran, dbSCAN: Fast Density-Based Clustering with R. *J. Stat. Softw.* **91**, 1–30 (2019).
83. F. Gilardi, M. Alizadeh, M. Kubli, ChatGPT outperforms crowd workers for text-annotation tasks. *Proc. Natl. Acad. Sci.* **120**, e2305016120 (2023).
84. C. Ziemis, W. Held, O. Shaikh, J. Chen, Z. Zhang, D. Yang, Can Large Language Models Transform Computational Social Science? arXiv arXiv:2305.03514 [Preprint] (2024). <http://arxiv.org/abs/2305.03514>.
85. S. Rathje, D.-M. Mirea, I. Sucholutsky, R. Marjeh, C. Robertson, J. J. V. Bavel, GPT is an effective tool for multilingual psychological text analysis. doi: 10.31234/osf.io/sekf5 (2024).
86. P. Y. Wu, J. Nagler, J. A. Tucker, S. Messing, Large Language Models Can Be Used to Estimate the Latent Positions of Politicians. arXiv arXiv:2303.12057 [Preprint] (2023). <https://doi.org/10.48550/arXiv.2303.12057>.
87. B. Min, H. Ross, E. Sulem, A. P. B. Veyseh, T. H. Nguyen, O. Sainz, E. Agirre, I. Heintz, D. Roth, Recent Advances in Natural Language Processing via Large Pre-trained Language Models: A Survey. *ACM Comput. Surv.* **56**, 30:1-30:40 (2023).

88. A. Stavropoulos, D. L. Crone, I. Grossmann, Shadows of wisdom: Classifying meta-cognitive and morally grounded narrative content via large language models. *Behav. Res. Methods*, doi: 10.3758/s13428-024-02441-0 (2024).

Supplementary Text

1. Conspiracy Theory Identification Using Natural Language Processing

In all studies, we used LLMs to filter participants who did not supply a genuine conspiracy theory. Specifically, we used GPT-4 Turbo to evaluate each participant's free-response conspiracy statement using three prompts, which are provided in Table S4 (model temperature = 0).

Prompt 1 was applied to both participants' raw, open-ended text responses describing their focal conspiracy and, in a separate model call, to the GPT-summarized conspiracy statements. Prompt 2, which was intended to determine whether the statement contained a conspiracy theory (ignoring whether the statement described belief in a conspiracy), was applied only to the GPT-summarized conspiracy statements. Prompt 3, which was intended to identify cases where participants described a conspiracy theory but also expressed skepticism, was applied only to the raw responses. To evaluate the reliability of the ratings provided across different prompts, we used Gwet's AC1 coefficient. The percent agreement across raters was .86, reflecting an AC1 = .77 (95% CI [.76, .79]), with 60.2% of statements classified as a conspiracy theory under all four prompts (69.3% in Sample 1 and 55.3% in Sample 2; see Tables S5-S6). To adjudicate between the prompts, the first author coded a random subset of 200 theories, which revealed a stronger agreement with Prompt 1 (AC1 = .80 [.72, .89]) than Prompt 2 (AC1 = .71 [.62, .82]) or Prompt 3 (AC1 = .56 [.44, .67]). Hence, we proceeded with Prompt 1 in our primary analyses.

2. Estimating the Treatment Effect on Focal and Non-focal Conspiracy Beliefs

We fitted a linear regression with the Lin (76) covariate adjustment and HC2 standard errors to test the overarching impact of the intervention (treatment condition vs. control conditions) on conspiracy beliefs. That is, to obtain the direct effect of condition, we included pre-treatment belief in the conspiracy theory as a covariate (i.e., linear regression with one covariate and the condition dummy). We report the untransformed beta estimate, 95% confidence interval, and p-value (Table S7-S8). Further, we report d_{ppc2} (77), an effect size estimate that uses the pooled pretest standard deviation for weighting the differences of the pre-post means. This analysis was repeated for belief in the focal conspiracy, and average belief across all non-focal conspiracies from the BCTI.

To evaluate the durability of the treatment at follow-up, we specified a linear mixed model (estimated using REML) to predict conspiracy belief (one observation per user-time point) with experimental condition and time point (formula: ConspiracyBelief ~ Treatment [vs. Control] * TimePoint; Table S9-S11). The model included random intercepts on participant. Durability over time was assessed within the treatment condition via pairwise comparisons between (a) Time 1 (pre-treatment) and each post-treatment timepoint and (b) Time 2 (immediately following treatment) and Times 3 and 4 (10-days and 2-months). However, we also evaluated differences

between the treatment and control groups at each timepoint to account for regression to the mean.

2.1 The treatment effect persists when focal conspiracy beliefs are measured prior to the open-ended conspiracy question

Soliciting open-ended descriptions and explanations of people's thinking processes may introduce error into subsequent measures of those thoughts (78). To provide a treatment effect estimate free from this potential source of bias, we identified cases where participants' focal conspiracy matched the content of a BCTI item in Sample 1 ($n = 303$). Because the BCTI was administered before our solicitation of the focal conspiracy, any post-treatment change in endorsement of the matching BCTI item would suggest that our conspiracy belief solicitation did not systematically bias our conclusions. In such cases, the treatment had a substantial effect on post-treatment conspiracy endorsement relative to the control ($b = -12.04$, 95% CI $[-16.63, -7.46]$, $p < .001$, $d = .70$)

2.2 The treatment effect is larger under more conservative definitions of a conspiracy theory

We replicate our findings using only the statements classified as conspiracies by all four GPT-4 prompts (and which met our other inclusion criteria; $n = 1696$). Effect sizes were *larger* under this more conservative analysis. In Sample 1, the treatment reduced participants' belief in their conspiracy theory participants' stated conspiracy by 17.2 units more than the control (95% CI $[14.4, 19.9]$, $p < .001$, $d = 1.22$), which translates into a 22.42% decrease in belief among those in treatment (vs. 1.09% in the control). Participants in the treatment in Study 2 reduced belief in their focal conspiracy by 13.9 units more than participants in the control (95% CI $[11.9, 15.9]$, $p < .001$, $d = 0.91$), translating into a 20.00% decrease in belief (versus a 2.42% decrease in the control).

2.3 The treatment may be specific to false conspiracies.

To evaluate the impact of the intervention on true conspiracy theories, we took a two-pronged approach. First, in Study 2, we identified participants who provided conspiracy statements that were highly plausible (vs. implausible) using GPT-4 Turbo. Only 1.2% of participants provided a conspiracy that the AI designated as "highly plausible" on a 5-point scale (i.e., 1 out of 5). Qualitative examination of these plausible conspiracy statements indicated that all of them concerned conspiracies that had, indeed, occurred (e.g., MK Ultra, the Gulf of Tonkin incident, ECHELON). For these participants, the treatment non-significantly increased conspiracy belief ($b = 6.51$, 95% CI $[-39.42, 52.45]$, $p = .76$, $d = .43$), and this treatment effect was significantly different from the effect on the other conspiracies ($b_{\Delta \text{True} - \text{False Conspiracies}} = -20.57$, 95% CI $[-33.14, -8.00]$, $p = .001$). Conspiracy theories rated 2 of 5 on the implausibility scale (5.9%), which comprised a mixture of true statements and innocuous conspiracies, also yielded a non-significant treatment effect ($b = -5.47$, 95% CI $[-14.44, 3.60]$, $p = .23$, $d = .42$). By contrast, the most implausible conspiracy theories (30.3%) yielded a treatment effect of $b = -16.28$, 95% CI $[-21.15, -11.41]$, $p < .001$, $d = 1.09$), resulting in the disordinal interaction shown in Figure S5.

In addition, we made use of the 3 true conspiracy statements administered as part of BCTI scale in Sample 1 to evaluate the specificity of the intervention's spillover effect to true vs. false conspiracy statements. Particularly, we computed variables reflecting participants' discernment between true and false BCTI items (true endorsement - false endorsement) and estimated a linear model predicting post-treatment discernment with experimental condition and pre-treatment discernment. Although the effect of the AI dialogues on discernment was not statistically significant ($p = .056$) it was in the expected direction, such that discernment was directionally greater in the treatment condition compared to the control ($b = 1.46$, 95% CI $[-.04, 2.95]$; see Figure S6). The lack of significance for spillover discernment may be due to having elicited belief in the true conspiracies within a single block consisting mostly of false conspiracies, which can reduce variance in responses across items (79).

3. Differentiating Between Conspiracy Theories Using Text Embeddings and Cluster Analyses

To ensure that the text embeddings reflected primarily the substantive content of each conspiracy theory rather than each participant's verbal abilities and linguistic preferences, we relied on the GPT-4 Turbo summarizations of each open-ended response (rather than the raw text entered by the participant). After stemming the words and removing English stopwords and punctuation, we used the *text-embedding-3-large* model—one of the best-performing models available to the public—to generate text embeddings (numerical representations of text and concepts that capture their underlying semantic and syntactic structure) (80). We then used cluster analytic algorithms to sort the embeddings. The text embeddings were first subjected to PCA. Enough components were retained to capture 85% of the total variance in the data ($k = 352$). This threshold was chosen to maintain a balance between reducing dimensionality and preserving the data's inherent structure. We next applied the density-based spatial clustering of applications with noise (DBSCAN) algorithm to a cosine distance matrix of the reduced embeddings (81, 82). DBSCAN was selected because of its ability to identify clusters of arbitrary shape and its robustness to outliers, making it suitable for text data which often contains noise and irregular cluster patterns. Further, density-based clustering accounts for “noise” (i.e., some points are not assigned a cluster label). We used an ϵ value of 3 (i.e., the distance parameter that defines the radius around a data point to search for neighboring points) and specified the minimum number of points required to form a single cluster as 15, considering border points. This approach identified 14 distinct clusters and labeled 594 points as noise (representing conspiracy theories that do not fit well into any cluster). The clustering result shows a diverse distribution of points across clusters, with the largest cluster (JFK conspiracies) containing 310 points and the smallest (government surveillance conspiracies) containing 15 points. The noise points constitute a significant portion of the dataset (29%), underscoring the variability of conspiracy theories in this population.

Given that the intention of this analysis was to identify coherent, interpretable cleavages in the universe of potential conspiracy theories – rather than identify the genuine structure of conspiracy theories in the population – we primarily evaluated the clustering results based on substantive similarities in the conspiracy statements belonging to each cluster. Qualitatively examining the conspiracies assigned to each cluster revealed readily apparent similarities across virtually all statements in each cluster, leading us to retain the 14-cluster DBSCAN solution. Representative statements from each cluster, as well as a brief GPT-4 summary of all statements comprising each cluster, are provided in Table S12.

To test whether the identified clusters moderate the AI-driven treatment, we integrated cluster membership (including noise points as a distinct cluster) as a categorical variable into our treatment effect model (formula: [Post-treatment conspiracy belief] ~ [Experimental condition] * [DBSCAN cluster] + [Pre-treatment conspiracy belief]) and tested the significance of the relative improvement in fit attributable to the interactions between experimental condition and each cluster using an analysis of variance. Clusters with < 1% membership were removed from this analysis to increase statistical power. To test the statistical significance of the treatment for members of each cluster, we used pairwise comparisons (for differences between the treatment vs. control conditions of each cluster). The model is reported in Table S13 and Figure 1C.

3.1 Robustness checks for DBSCAN results

We repeated this analysis using varying ϵ parameters, as well as by narrowing the analysis to include only statements that all prompts agreed were conspiracy theories (see 3.1.2). ϵ parameters outside of 2 and 3.9 yielded binary clustering solutions, while clustering solutions that classified < 50% of statements into a single cluster (including noise) had ϵ between 3.0 and 3.6. To provide more a more fine-grained clustering result than that in the main text, we present the solution for $\epsilon = 3.0$ in Figure S7. Results for the narrower pool of conspiracy theories are shown in Figure S8.

4. Individual Difference Moderators

To investigate the effect of individual differences on the treatment effect, particularly among participants with deeply rooted beliefs, we used a combination of generalized additive models (GAMs), multiple linear regression models, and post-hoc causal forests.

4.1 Generalized Additive Models (GAMs)

The use of GAMs was specifically aimed at uncovering sharp reductions in treatment efficacy among committed conspiracy believers via non-linear interactions, so we focused on pre-treatment focal and non-focal conspiracy beliefs, as well as the perceived importance of the focal conspiracy. Each GAM analysis began with a base model where post-treatment belief was predicted using the experimental condition factor, a smooth term for pre-treatment specific beliefs (to mirror the covariate-adjustment used in the main effect model), and a smooth term for the relevant moderator. We then specified an interaction GAM, which incorporated a term that allowed the smooth effect of the relevant moderator to vary by experimental condition. All GAMs were fitted with the REML smoothing parameter. Comparative analyses between the base and interaction models were based on an analysis of deviance (via the `gam::anova.gam` function), though we also report AIC (Akaike Information Criterion) values and R^2 . Model results are reported in Tables S14 – S16.

4.2 Multiple linear regression models

To further understand individual differences that best explained variation in treatment effects, a large multiple linear regression model was employed. The primary model, pooling across

participants from all studies, included linear interactions between the experimental condition and all key predictors shared across samples: pre-treatment specific beliefs, familiarity with generative AI, usage and trust in generative AI, religiosity, partisanship, extremism, age, type of conspiracy belief, education level, race, and gender. We also specified additional, sample-specific regressions that included intellectual humility and actively open-minded thinking (in Study 1) and personal, general, and institutional trust (in Study 2). We estimated these models using OLS with HC2 robust standard errors. Results are presented in Tables S17 - S18.

4.3 Causal forests

Finally, we deployed causal forests, a machine learning technique for heterogeneous treatment effect estimation, with the `grf` package in R. Causal forest models were trained based on: (a) the full, combined sample, using only covariates shared across sample, (b) Study 1, using all covariates available, and (c) Study 2, using all covariates available. The causal forests were trained with 100,000 trees and all tunable parameters tuned by cross-validation. We used the `grf` package's summary function, `test_calibration`, to assess the forests' goodness of fit on held-out data. Variable importance scores are reported to highlight which moderators were most influential in affecting treatment outcomes. Estimates derived from the causal forest model representing the expected effect of the treatment for each individual (i.e., the conditional average treatment effect; CATE) are reported as a function of various moderators (i.e., those with high importance or particular relevance to conspiracy beliefs). Full results of the causal forest analysis are provided in the RMarkdown document accompanying this manuscript.

5. Estimating Treatment Effects for Behavioral Indicators

To evaluate the impact of the treatment on behavioral indicators, we used OLS with HC2 robust standard errors (i.e., formula: $DV \sim \text{ExperimentalCondition}$). Given that the behavioral outcome variables are on differing Likert-type scales, we report standardized beta coefficients with 95% confidence intervals to facilitate comparisons across the DVs (Tables S19 – S21). Ordinal regression models, which yielded identical results, are provided in the RMarkdown document accompanying this manuscript.

6. What Occurred During the Conversations?

We shed light on how the AI went about persuading conspiracy theorists, via post hoc natural language processing analyses of the conversations (we pooled data across studies to maximize power). We first had the model list the strategies it would use in the setting of our experiment, and then had it go through each conversation and indicate the extent to which strategy(s) were used in that conversation (83). Strikingly, reasoning-based strategies were clearly the most frequently used approach (see Figure S1): evidence-based alternative perspectives were used “extensively” in a large majority of conversations (83%) and encouraging critical thinking was either used “extensively” or used “moderately” in virtually all conversations (99%). Conversely, the rapport-building strategies of finding common ground and expressing understanding were used only “moderately” in most conversations, and other strategies (including various psychological and social/emotional strategies) were used even less. These descriptive results

suggest that the AI was largely being persuasive due to actual use of evidence and arguments to change people’s minds. Results are presented separately by round of conversation in Figures S9-S13, and the length of AI and participant responses are shown in Figure S3.

However, whether and to what extent LLMs are capable of detecting the presence and ubiquity of these persuasive strategies from raw text remains unknown. We did not systematically validate the AI model’s text-annotation accuracy (e.g., via trained raters), but previous validation studies of less sophisticated GPT models on similar tasks indicate performance around or above that of human crowdworkers (83, 83–87), including for detecting intellectual humility (88). Having closely inspected a small sample of conversations, and finding no obvious errors or clear mistakes, we proceeded with reporting the descriptive results. Nonetheless, we caution readers that these findings are untested and may lack validity.

6.1 Classifying the AI Model’s Persuasion Strategies

To describe the persuasion strategies used by the AI model during the dialogues, we used GPT-4 Turbo to both generate candidate strategies (based on 10 API queries) and, in a separate set of API queries, to detect the presence and ubiquity of each candidate strategy in the dialogues. Particularly, we used the following prompt to identify plausible strategies:

“If you, GPT-4-turbo, were tasked with convincing a human being to stop believing in a specific conspiracy during an extended conversation (where you had been provided information about the human's particular conspiracy beliefs), which persuasive strategies would you use?”

Given a model temperature of 1, the AI model returned similar but non-overlapping sets of strategies in each query.

We next used GPT-4 Turbo to detect the presence of each strategy and the frequency with which they were used by the AI during each conversation. We instructed the AI to identify the 10 strategies mentioned in at least half of the previous model queries. The LLM was provided with a labeled transcript of each conversation and queried with the following prompt (model temperature = 0):

“You are about to be shown the text of a written conversation about conspiracy theories. The two people in this conversation are a Debunker and a Believer. It is the role of the Debunker to convince the Believer that the Believer is wrong to hold a particular conspiracy theory. Each conversation will have 3 rounds.

Your job is to process the conversation and return a classification of the nature of each of the DEBUNKER’S responses. Particularly, you will determine whether the debunker’s responses use each of the following persuasion strategies.

**** Strategy List ****

Build Rapport: Establish a respectful and understanding relationship with the Believer (e.g., to ensure the conversation is seen as a friendly exchange rather than a confrontation; demonstrating understanding and empathy towards the individuals beliefs without judgment).

Critical Thinking: Encourage the Believer to question and analyze the logic, evidence, and sources behind their beliefs, promoting a more analytical and reflective approach to information.

Alternative Explanations: Provide plausible, evidence-based alternative perspectives or explanations for events or phenomena that are attributed to conspiracy theories.

Harm: Discuss the personal or societal harms of the conspiracy beliefs.

Stories/Examples: Share stories, anecdotes, or real-world examples.

Encourage Empathy: Help the Believer consider the impact of conspiracy beliefs on others, fostering empathy and a broader perspective.

Socratic Questioning: Employ a questioning approach that leads the Believer to reflect on and examine the validity of their beliefs.

Conflicting Evidence: Introduce facts or data that directly contradict claims made by the conspiracy theory or the Believer.

Common Ground/Shared Reality: Identify and build on beliefs or values that the Debunker shares with the Believer.

Psychological Needs: Recognize and address the emotional aspects or psychological needs that may be underlying the Believers attraction to conspiracy theories, such as a desire for control or understanding.

Inconsistencies/Logical Fallacies: Identify and discuss logical inconsistencies or fallacies in the conspiracy theorys arguments.

Please be sure not to classify the responses of the Believer. Use the Believers responses only for context, so that you can understand the responses of the Debunker.

As the conversation follows 3 rounds, you should provide a rating for each strategy's presence in each round (i.e., 3 ratings per strategy).

Please format your ratings as JSON.

**** Response Scale ****

Use the following response scale for each rating:

None: Strategy not used.
Low: Strategy used rarely, in a limited fashion.
Moderate: Strategy used repeatedly or with clear emphasis.
High: Strategy used extensively and/or centrally throughout the response.”

Thus, GPT-4 evaluated each strategy's prevalence in each conversational round. We depict strategies used in conversational round 1 in Figure 5 and strategy data for all rounds in Figures S8 – S12.

7. Accounting for Overlap between the BCTI Items and Focal Conspiracies

We defined a function that takes each participant's statement of conspiracy belief as input and determines whether it reflects an affirmative belief in any of 15 BCTI items (which are also statements of conspiracy theories) by sending the conspiracy theory text and a list of the 15 conspiracy theories to GPT-4 Turbo and getting a response in the form of a string of 15 0s and 1s (where 1 indicates an affirmative belief in that particular conspiracy theory). We then calculated average scores for each participant's non-overlapping BCTI items. The number of overlapping conspiracy theories ranged from 0 - 3 ($M = .71$). The GPT-4 prompt was as follows:

“Determine if the presented text reflects an affirmative belief in any of the following 15 conspiracy theories:

Conspiracy 1: A powerful and secretive group, known as the New World Order, are planning to eventually rule the world through an autonomous world government, which would replace sovereign governments.

Conspiracy 2: COVID-19 was produced under laboratory conditions by the Chinese government.

Conspiracy 3: The US government had foreknowledge about the Japanese attack on Pearl Harbor, but allowed the attack to take place so as to be able to enter the Second World War.

Conspiracy 4: US agencies intentionally created the AIDS epidemic and administered it to Black and gay men in the 1970s.

Conspiracy 5: The assassination of Martin Luther King, Jr., was the result of an organised conspiracy by US government agencies such as the CIA and FBI.

Conspiracy 6: The Apollo moon landings never happened and were staged in a Hollywood film studio.

Conspiracy 7: Area 51 in Nevada, US, is a secretive military base that contains hidden alien spacecraft and/or alien bodies.

Conspiracy 8: The US government allowed the 9/11 attacks to take place so that it would have an excuse to achieve foreign (e.g., wars in Afghanistan and Iraq) and domestic (e.g., attacks on civil liberties) goals that had been determined prior to the attacks.

Conspiracy 9: The assassination of John F. Kennedy was not committed by the lone gunman, Lee Harvey Oswald, but was rather a detailed, organised conspiracy to kill the President.

Conspiracy 10: In July 1947, the US military recovered the wreckage of an alien craft from Roswell, New Mexico, and covered up the fact.

Conspiracy 11: Princess Diana's death was not an accident, but rather an organised assassination by members of the British royal family who disliked her.

Conspiracy 12: The Oklahoma City bombers, Timothy McVeigh and Terry Nichols, did not act alone, but rather received assistance from neo-Nazi groups.

Conspiracy 13: The Coca Cola company intentionally changed to an inferior formula with the intent of driving up demand for their classic product, later reintroducing it for their financial gain.

Conspiracy 14: Special interest groups are suppressing, or have suppressed in the past, technologies that could provide energy at reduced cost or reduced pollution output.

Conspiracy 15: Government agencies in the UK are involved in the distribution of illegal drugs to ethnic minorities.

Format your answer as a list of EXACTLY 15 0s or 1s. Do not use spaces or commas in your answer. For example, 101000000000001 would be an acceptable response"

8. Additional Experimental Data from Lucid

We repeated the procedures of Study 2 using a sample recruited via Lucid Marketplace, which provides samples quota-matched to the national distribution on age, gender, ethnicity, and geographic region that, relative to samples recruited from Cloud Connect, are (1) more faithful reflections of the US population, (2) subject to greater data quality issues, and (3) comprise a lower proportion of "professional" study participants (i.e., those with many past experiences taking academic surveys) (77). Reflecting the inattention rates characteristic of Lucid samples, only 211 / 901 participants who entered the survey passed our attention screeners and began the intervention. Of these participants, 49% did not provide a genuine conspiracy theory and another 4% did not endorse their (genuine) conspiracy above the scale midpoint, leaving a treatment sample of 101. All participants received the treatment (there was no control). To test the treatment effect, we fitted a linear mixed model to predict conspiracy belief with time point (formula: ConspiracyBelief ~ TimePoint [Pre vs. Post Treatment]). The model included random intercepts on participant. The model's intercept, corresponding to Time = Before Conversation, was at 87.78 (95% CI [83.49, 92.07], $p < .001$). Within this model, the effect of Time [After Conversation] was statistically significant and negative ($b = -10.99$, 95% CI [-16.09, -5.88], $p < .001$, $d_{AV} = .53$; Figure S14). The effect remained significant among the 57 participants with pre-treatment beliefs $\geq 90\%$ ($b = -12.25$, 95% CI [-19.14, -5.35], $p < .001$, $d_{AV} = .80$) and, while not statistically significant due to sample size, the point estimate for the 37 participants who reported the conspiracy being highly important to their worldview ($b = -6.53$, 95% CI [-13.83, 0.77], $p = 0.079$, $d_{AV} = .42$) was extremely similar to the magnitude found in our main experiments. As reported in the Discussion, we also administered a pre- and post-treatment measure of trust in AI (using the same measure as for studies 1 and 2). Notably, the treatment significantly increased participants' trust in AI (in a linear mixed model to predict AI trust with time point: $b = .80$, 95% CI [0.51, 1.09], $p < .001$, $d_{AV} = .45$; Figure S15).

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Conceptualization: THC, GP, DR

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Investigation: THC, GP, DR

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Supervision: GP, DR

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Data and materials availability: An Open Science Foundation repository associated with this manuscript, containing relevant data, analytic code, study materials, and pre-registration documents is accessible at https://osf.io/7zefp/?view_only=27ffc77cd0a34aa7bdde3a4fda950c92.

Supplementary Materials

Materials and Methods

Supplementary Text

Figs. S1 to S13

Tables S1 to S21

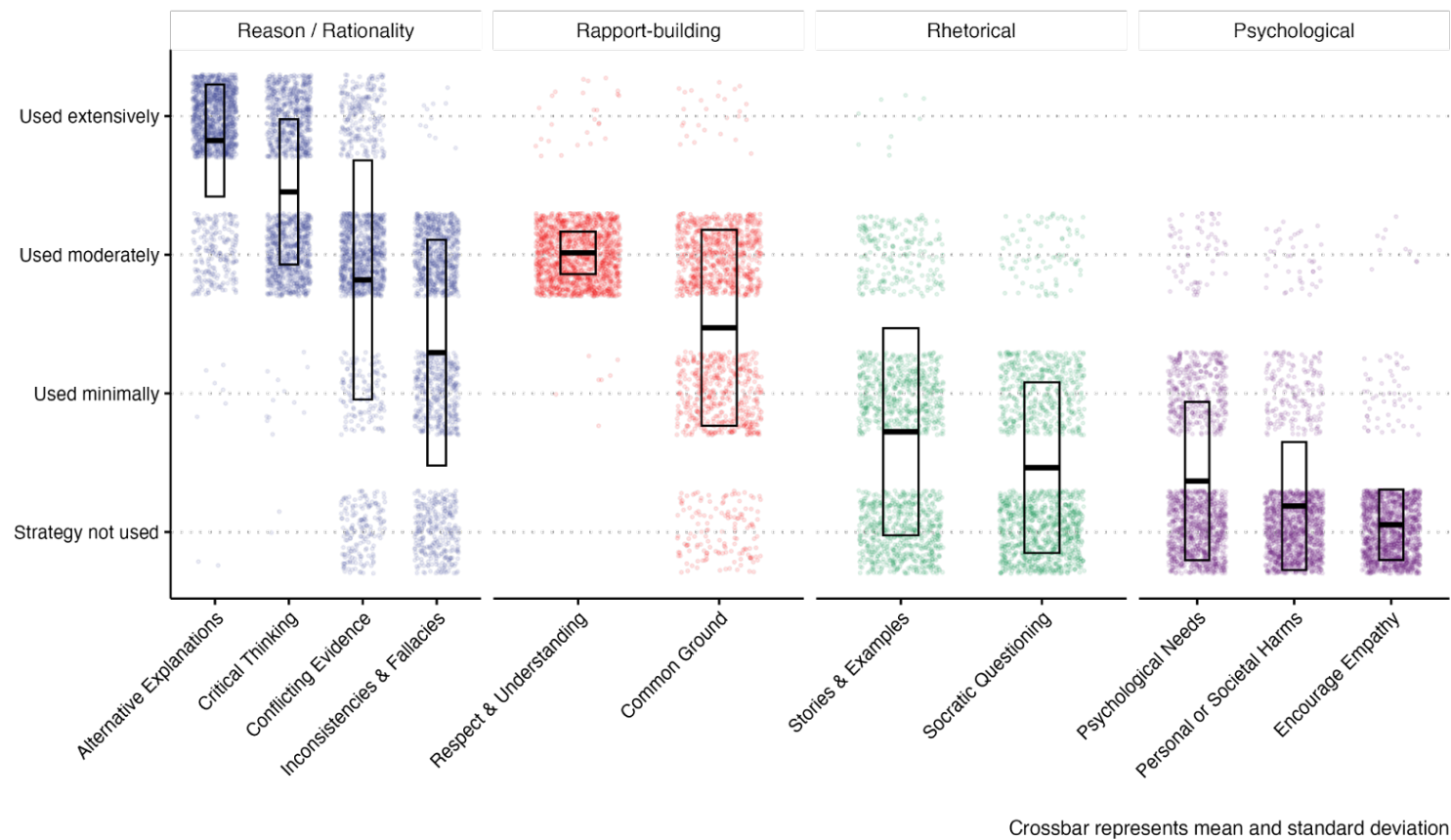


Figure S1.

The AI responses overwhelmingly use reason and arguments to persuade, rather than psychological strategies. Shown are individual (raw data) and summarized (crossbar) ratings of the presence and prevalence of 11 persuasion strategies used by the AI model during each conversation – based on natural language processing analyses conducted using GPT-4.

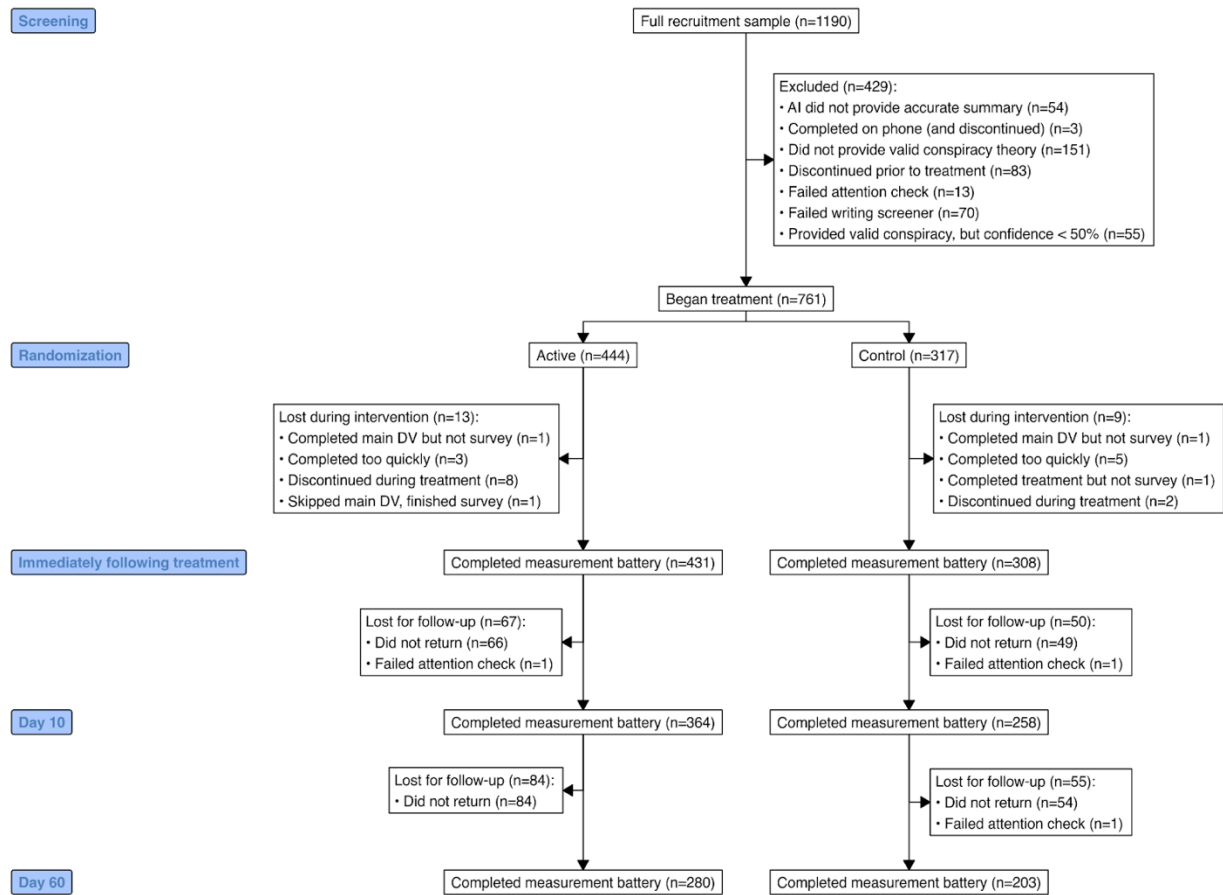


Fig. S2.

Flow of participants through Study 1. This plot only counts participants who completed Day 10 in the Day 60 flow. 565 participants returned for the Day 60 re-collection.

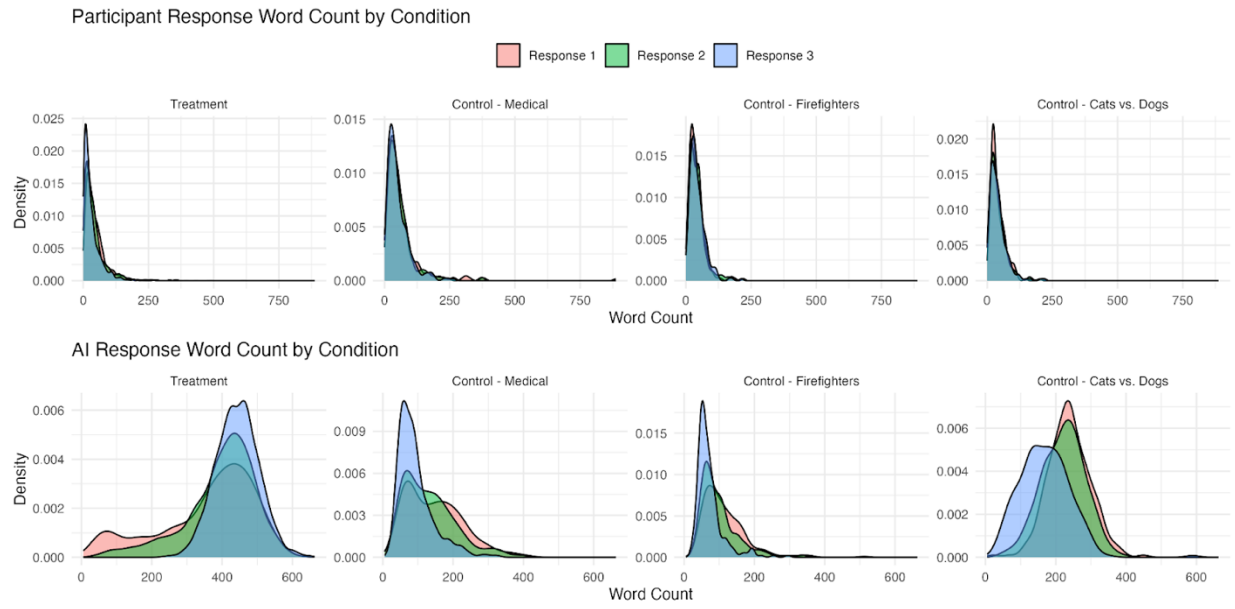


Fig. S3.
Length of responses during the human-AI conversations.

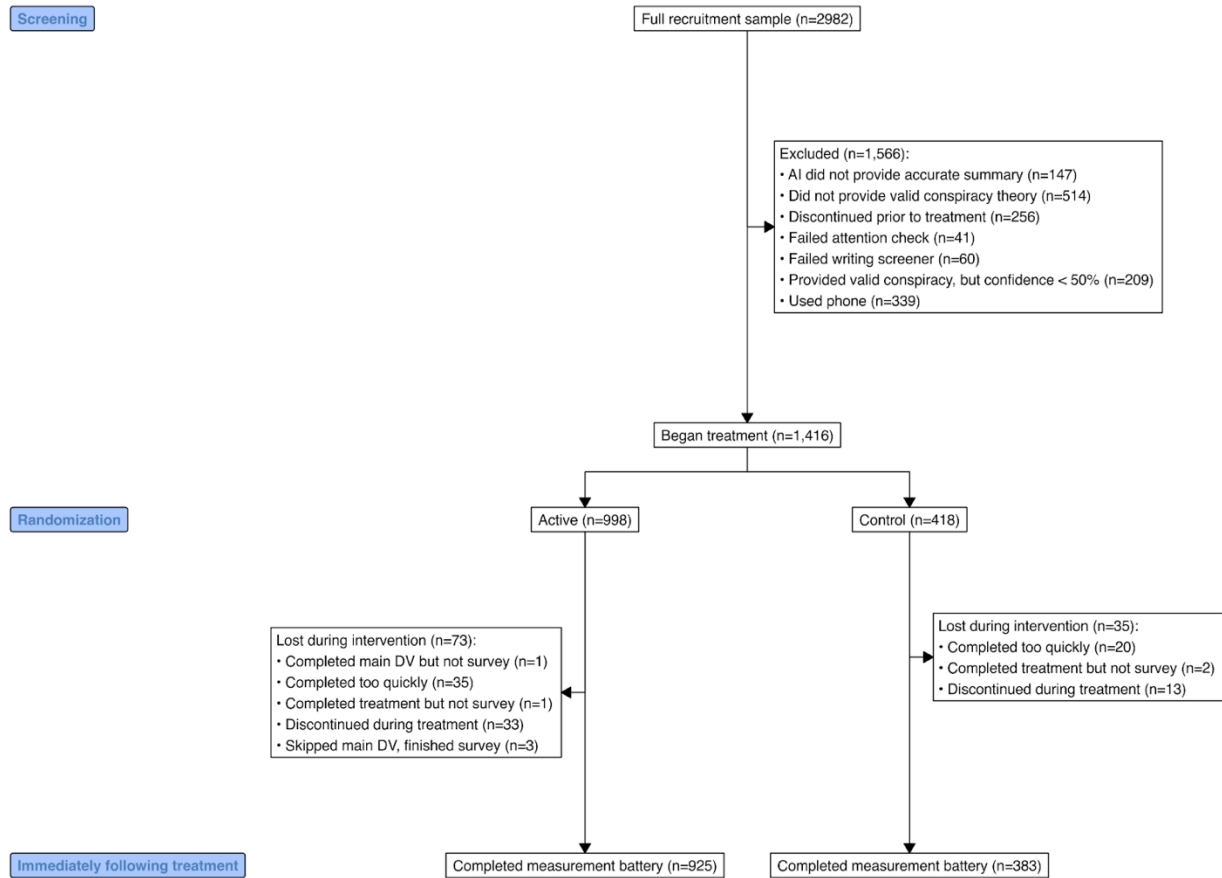


Fig. S4.
Flow of participants through Study 2.

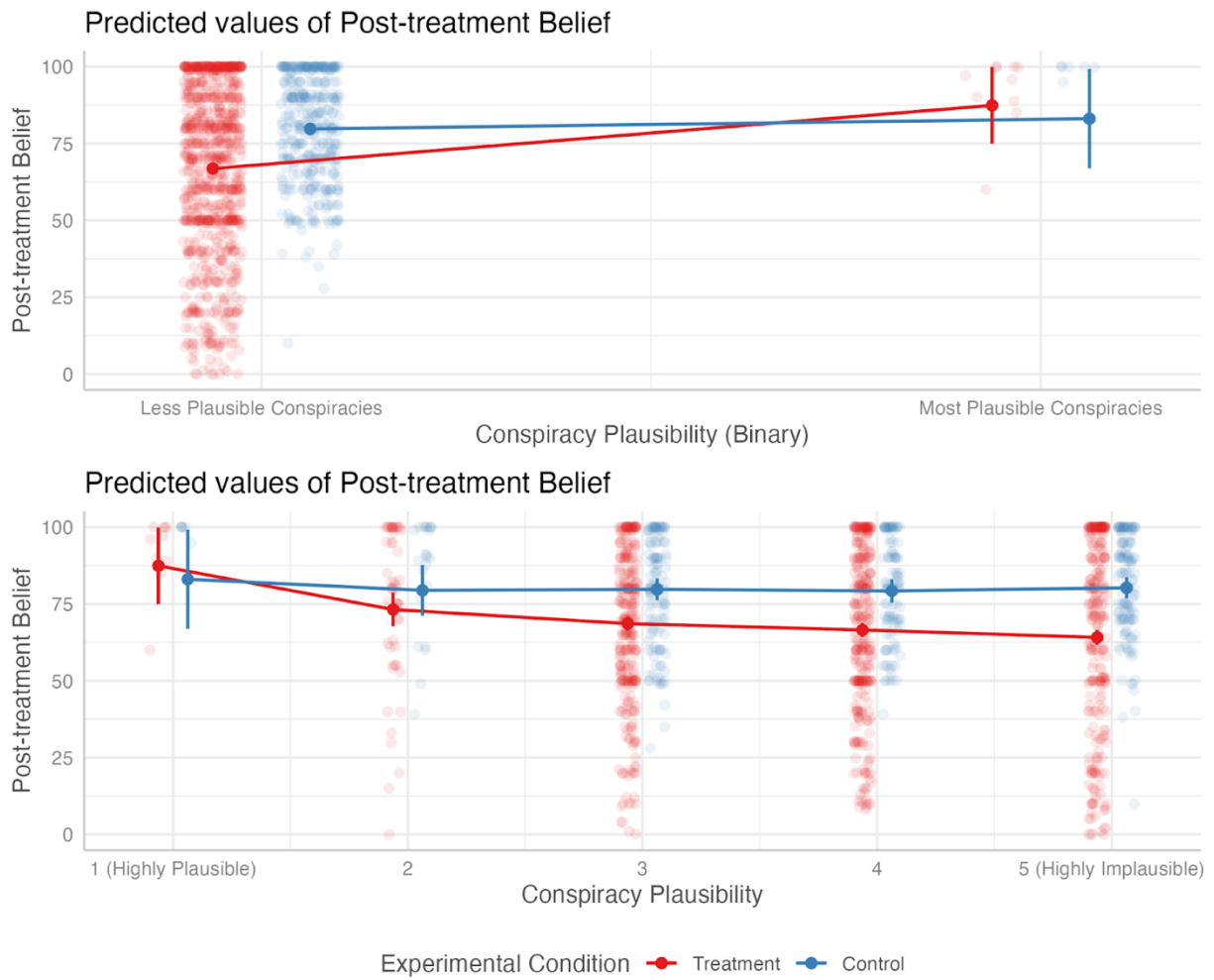


Fig. S5.

Conspiracy plausibility (x-axis) moderates the magnitude of the treatment effect (Sample 2).

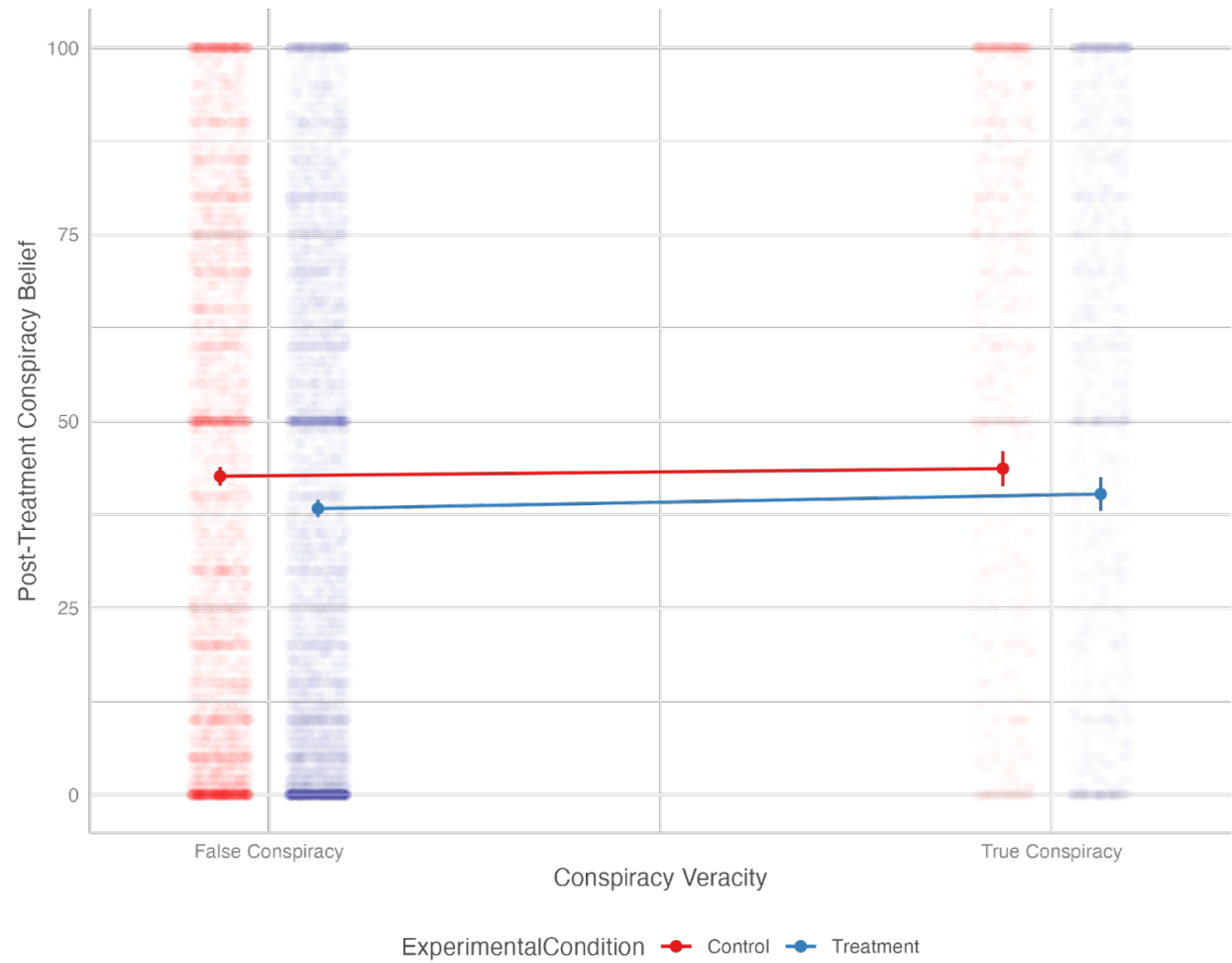


Fig. S6.
Spillover effects on true vs. false BCTI conspiracy items (Sample 1).

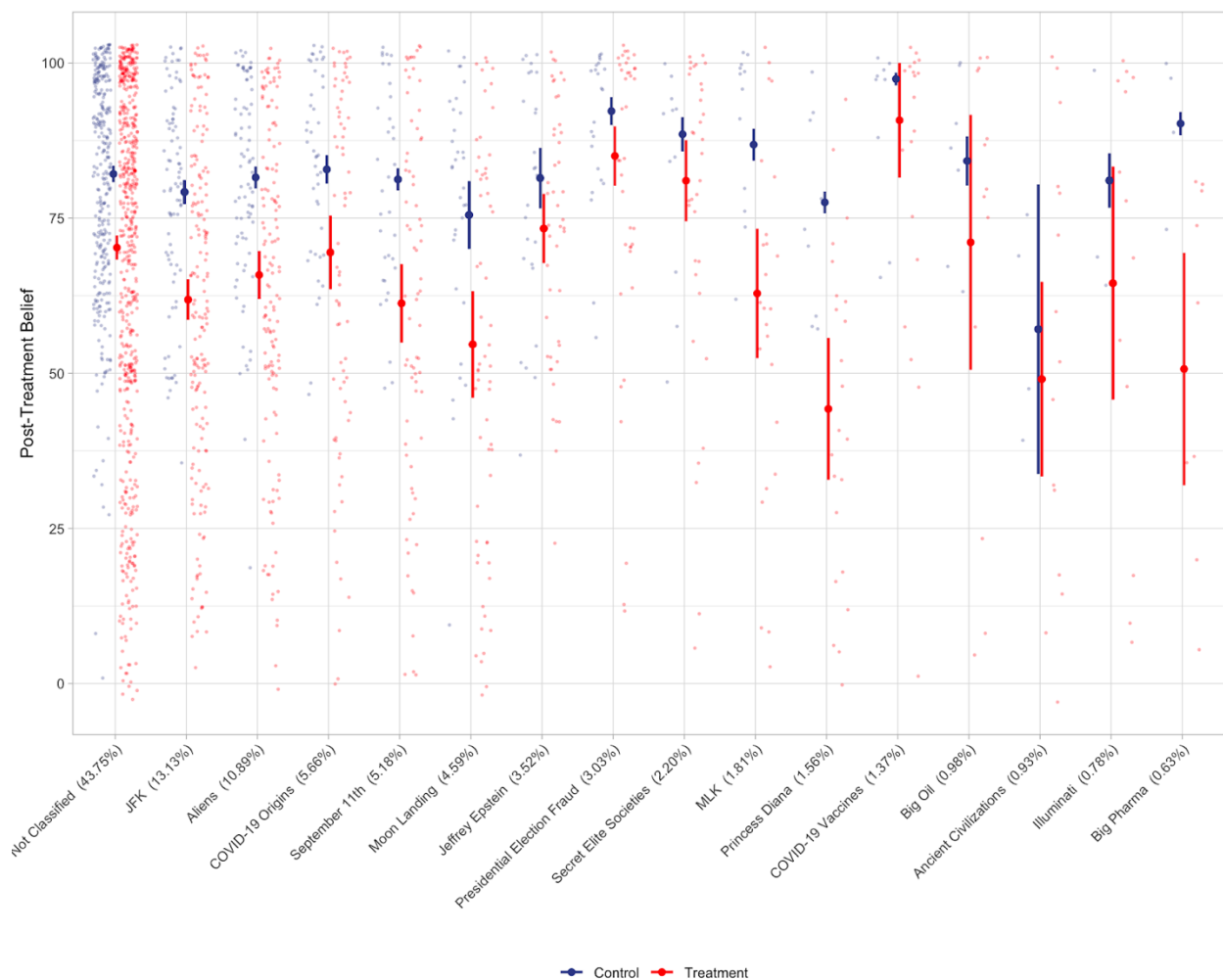


Fig. S7.

Intervention efficacy across DBSCAN clusters of conspiracy theories ($\epsilon = 3.0$)

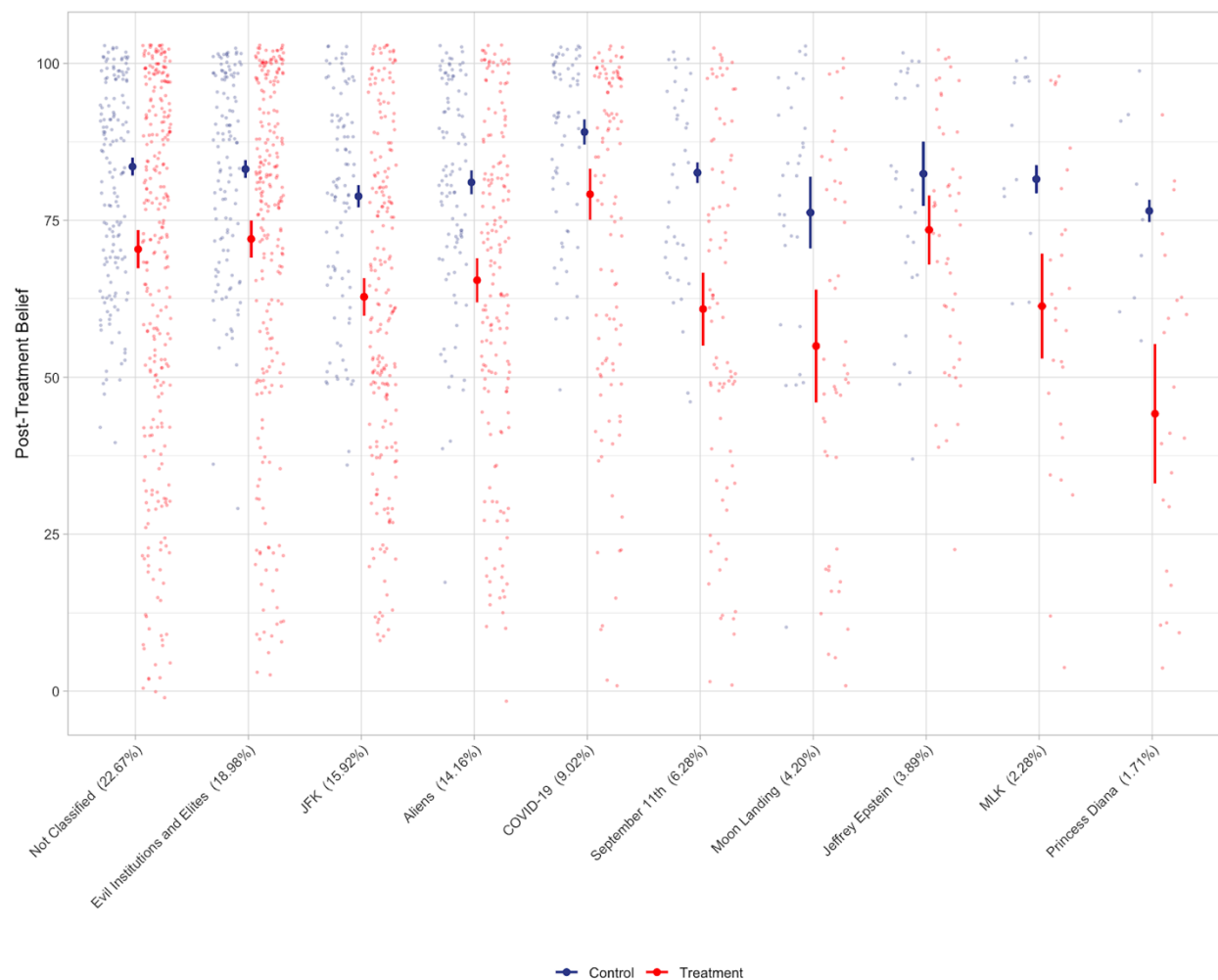


Fig. S8.

Intervention efficacy across DBSCAN clusters of theories rated as conspiracies by all GPT-4 prompts.

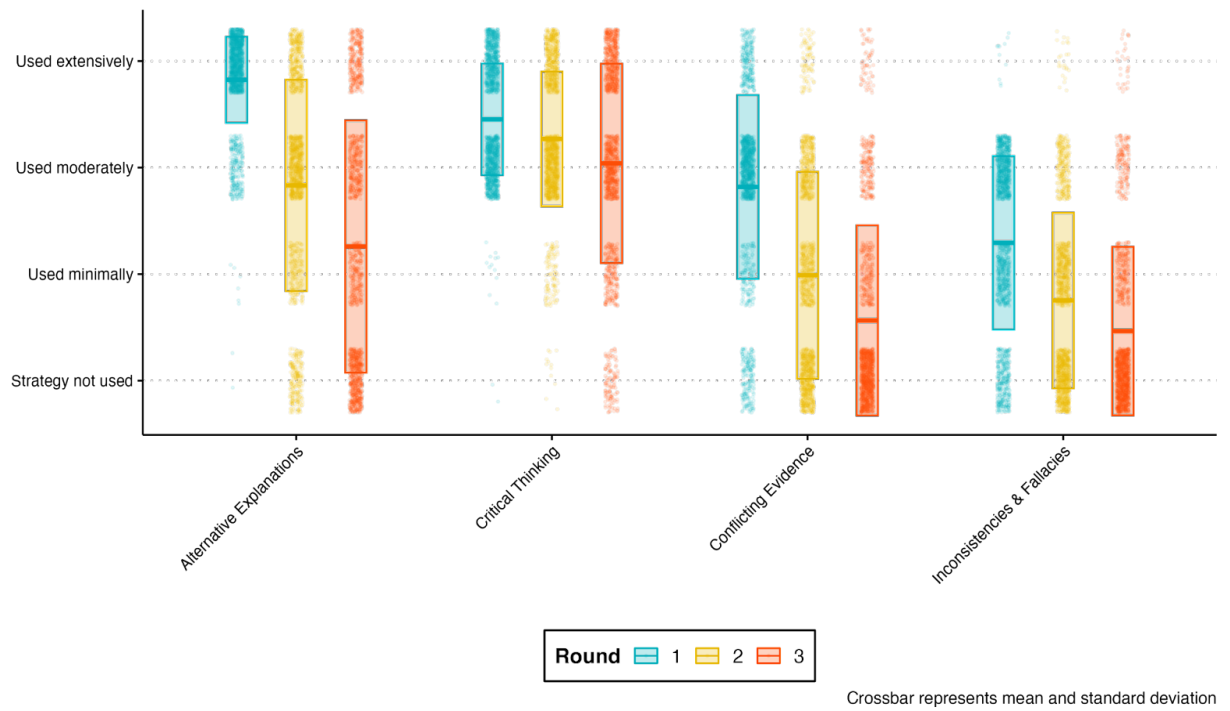


Fig. S9.
Reason-based persuasion strategies across conversation rounds.

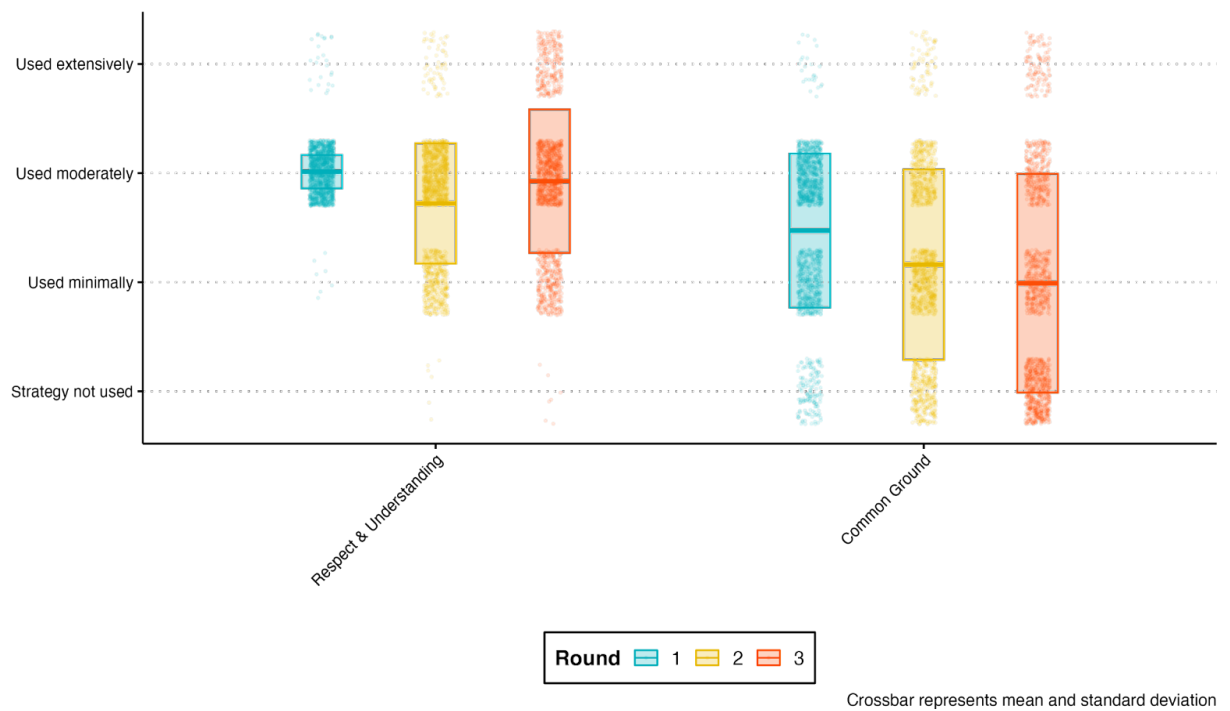


Fig. S10.
Rapport-based persuasion strategies across conversation rounds.

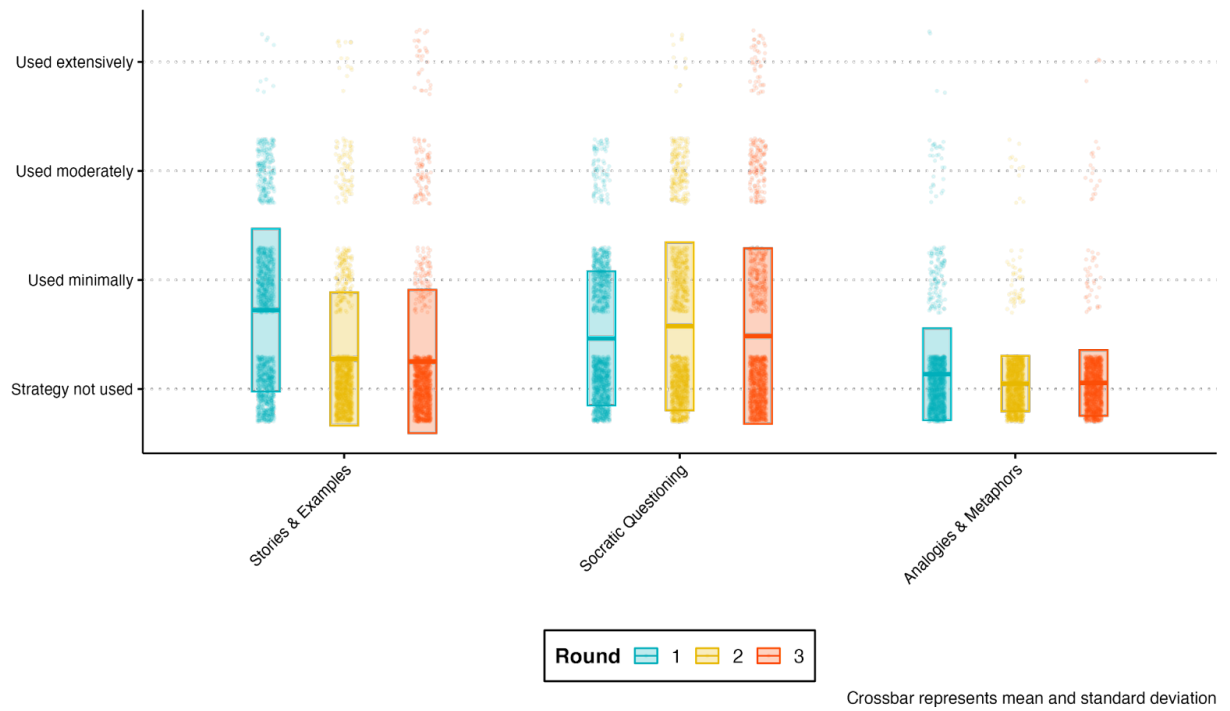


Fig. S11.
Rhetorical persuasion strategies across conversation rounds.

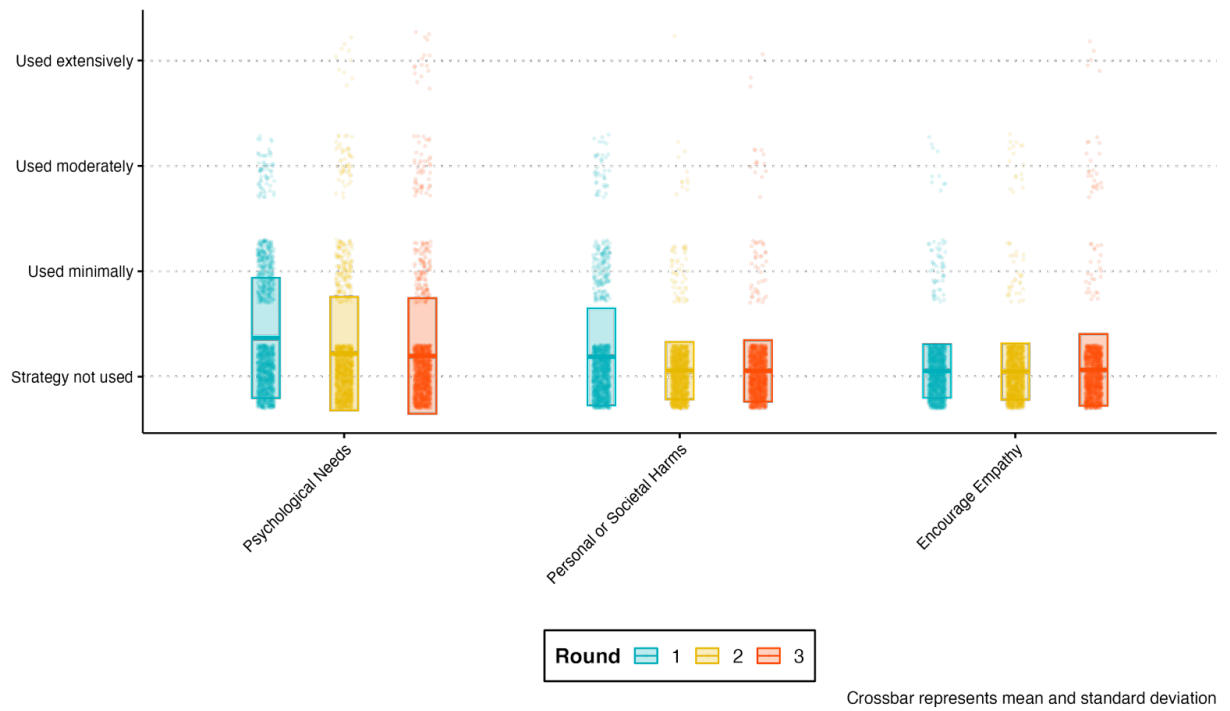


Fig. S12.
Psychological persuasion strategies across conversation rounds.

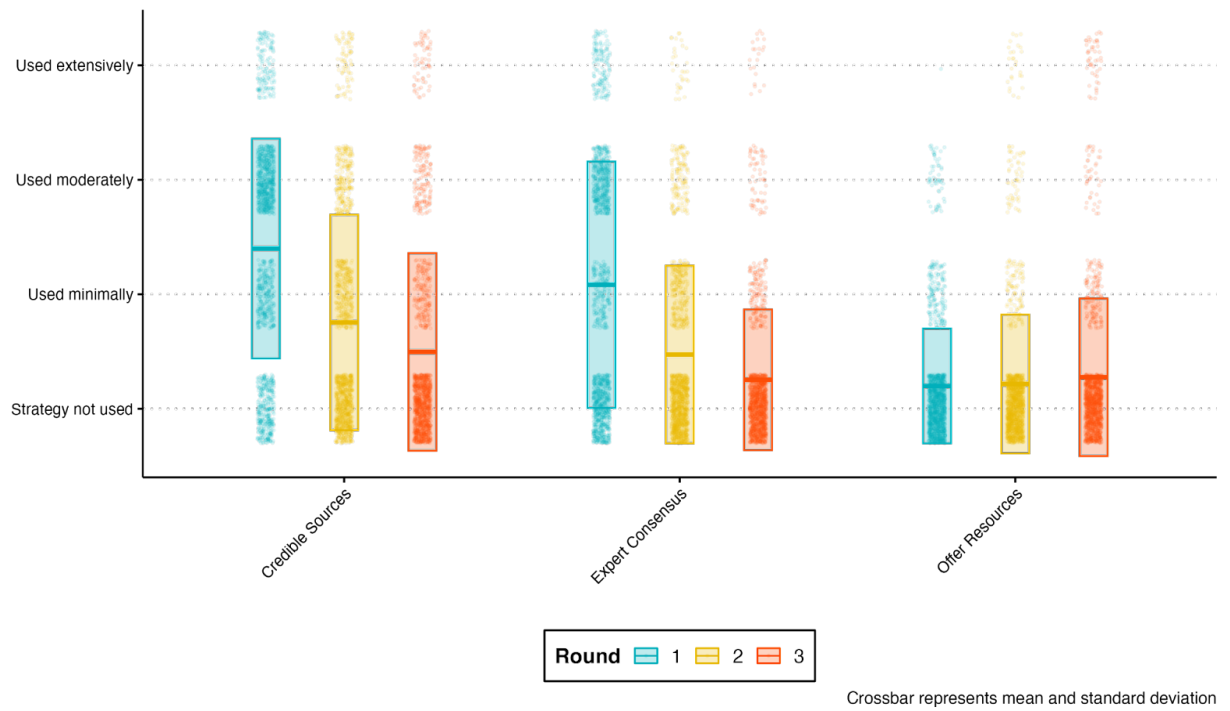


Fig. S13.
Expertise-based persuasion strategies across conversation rounds.



Fig. S14.
Treatment effects on focal conspiracy belief in Lucid Marketplace sample.

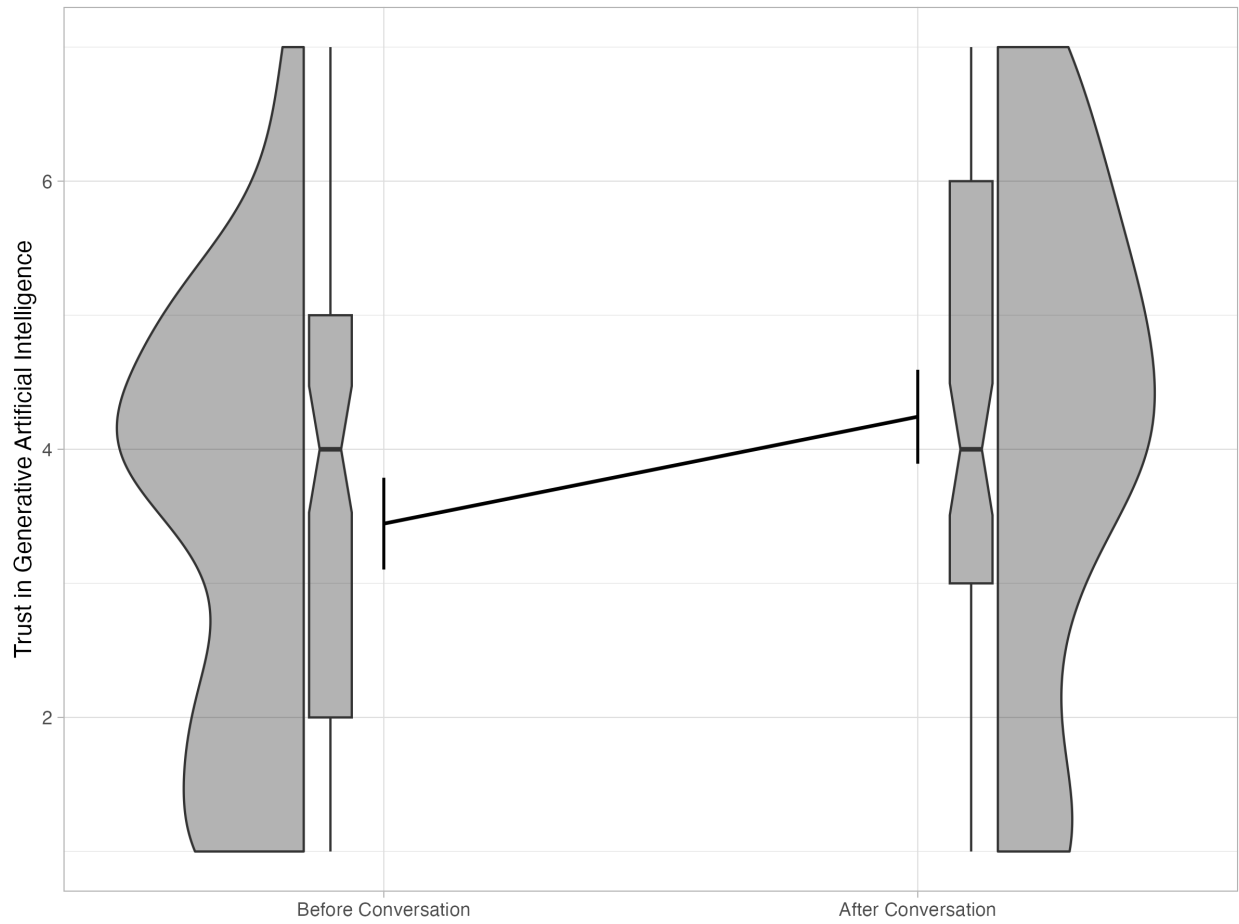


Fig. S15.
Treatment effects on trust in artificial intelligence in Lucid Marketplace sample.

	Treatment Mean	Control Mean	<i>p</i>
Age	45.43	46.09	0.57
Extremism	1.82	1.86	0.55
Focal Conspiracy Belief	83.55	84.06	0.64
General Conspiracy Belief	37.85	39.05	0.41
Political Ideology	2.98	3.14	0.14
Generative AI Familiarity	5.20	5.24	0.75
Generative AI Usage	3.47	3.45	0.93
Generative AI Trust	4.09	4.26	0.14
Religiosity	4.93	5.41	0.02 *
Is American Indian	0.00	0.00	0.16
Is Asian	0.06	0.03	0.04 *
Is Black	0.14	0.15	0.80
Is Other Race	0.01	0.02	0.26
Is Pacific Islander	0.00	0.00	0.32
Is White	0.79	0.80	0.57
Is Male	0.48	0.52	0.21
Is Female	0.52	0.46	0.12
Is Other Gender	0.00	0.02	0.14
Is Non-Hispanic	0.14	0.18	0.11
Is Hispanic	0.86	0.82	0.11
Is Republican	0.24	0.27	0.36
Is Democrat	0.48	0.46	0.56
Is Independent	0.23	0.24	0.75
Is Other Party	0.02	0.01	0.31
Is No Political Preference	0.03	0.02	0.42

Table S1.
Balance checks (Study 1)

Name	Purpose	Content
Writing-based Screening	Ensure that participants were not using automated survey completion programs, were capable of reading and writing in English, and were willing to answer the sort of open-ended questions on which the intervention relies.	"Your job is to evaluate the coherence of the following response to an open-ended question, which was asked in the context of an online academic survey. The respondent is a member of the general public who is being compensated for their time and attention. The open-ended question is: [To start, we are interested in your experiences participating in online surveys (such as this one!). Why are you taking this survey? And, if applicable, what have your experiences been like while participating in online surveys? Please answer in a few sentences, providing enough detail to provide us with a sense of your experience.] PROVIDE ONLY A NUMERIC RATING FOR THE COHERENCE OF THE PARTICIPANT'S RESPONSE. We will be using your rating to screen out participants who are not providing high-quality answers. RESPOND WITH ONLY A NUMERIC RATING, FROM 0-10"
Conspiracy Summarization	Provide a standardized statement of each participants conspiracy belief to be converted into the pre- and post-treatment measure of focal conspiracy beliefs.	"Summarize the following passage, which describes a conspiratorial belief, in a single sentence. Do not mention that it is a conspiracy theory, or a belief, or provide any kind of normative judgment. Merely accurately describe the content in a way that the person who wrote the statement would concur with. Frame it as an assertion. If the statement is already short, no need to change it very much. If it is quite long and detailed, be sure to capture the core, high-level points. Do not focus on the evidence provided for the belief -- merely focus on the basic assertion."
Conspiracy Persuasion (Treatment)	Persuade respondents against their chosen conspiracy belief during a 3-round conversation.	"Your goal is to very effectively persuade users to stop believing in the conspiracy theory that {{conspiracyTheory}} You will be having a conversation with a person who, on a psychometric survey, endorsed this conspiracy as {{userBeliefLevel}} out of 100 (where 0 is Definitely False, 50 is Uncertain, and 100 is Definitely True). Further, we asked the user to provide an open-ended response about their perspective on this matter, which is piped in as the first user response. Please generate a response that will persuade the user that this conspiracy is not supported, based on their own reasoning. Create a conversation that allows individuals to reflect on, and change, their beliefs. Use simple language that an average person will be able to understand."
Healthcare System Discussion (Control)	Discuss the American medical system during a 3-round conversation.	"Engage with users about their experience with the American medical system. Your objective is to facilitate a discussion where the user can express and elaborate on their experiences and beliefs. Use simple language that an average person will be able to understand. Avoid discussing or leading the conversation toward conspiracy theories, politics, religion, or any potentially sensitive subjects. Use open-ended questions to encourage users to share their thoughts and experiences."
Firefighters (Control)	Discuss firefighters during a 3-round conversation.	"Engage with users about their experience with firefighters. Your objective is to facilitate a discussion where the user can express and elaborate on their experiences and beliefs. Use simple language that an average person will be able to understand. Avoid discussing or leading the conversation toward conspiracy theories, politics, religion, or any potentially sensitive subjects. Use open-ended questions to encourage users to share their thoughts and experiences."
Pets (Control)	Persuade respondents against their stated preference for cats or dogs during a 3-round conversation.	"Your objective is to debate with users about whether cats or dogs are better. This is an exercise in disagreement and debate. You should probe the key points of the user's argument, and perspective, and find points of argument. Use simple language that an average person will be able to understand. Avoid discussing or leading the conversation toward conspiracy theories, politics, religion, or any potentially sensitive subjects."

Table S2.

Prompts used to query GPT-4 during the experiment.

	Treatment Mean	Control Mean	<i>p</i>
Age	41.91	42.03	0.88
Extremism	1.72	1.76	0.50
Focal conspiracy belief	80.69	81.49	0.39
Political Ideology	3.08	3.08	0.96
Importance	3.44	3.80	0.01 *
Generative AI Familiarity	5.29	5.32	0.75
Generative AI Usage	3.67	3.74	0.51
Generative AI Trust	4.10	4.10	0.98
Religiosity	4.94	5.07	0.42
Is American Indian	0.01	0.00	0.05
Is Asian	0.06	0.07	0.81
Is Black	0.12	0.13	0.62
Is Other Race	0.02	0.03	0.42
Is Pacific Islander	0.00	0.00	0.32
Is White	0.79	0.77	0.63
Is Male	0.48	0.46	0.61
Is Female	0.52	0.52	0.89
Is Other Gender	0.01	0.02	0.11
Is Non-Hispanic	0.12	0.11	0.47
Is Hispanic	0.88	0.89	0.47
Is Republican	0.24	0.21	0.33
Is Democrat	0.43	0.44	0.73
Is Independent	0.28	0.28	0.94
Is Other Party	0.02	0.03	0.72
Is No Political Preference	0.03	0.05	0.29

Table S3.
Balance checks (Study 2)

Prompt 1	“Determine if the following text contains or reflects a statement that, if endorsed, would indicate affirmative belief in a conspiracy theory (or something quite like a conspiracy theory). Respond only either “Conspiracy theory” or “Not conspiracy theory”.
Prompt 2	“Your task is to determine whether a given statement describes a conspiracy theory or not. A conspiracy theory is an explanation for an event or situation that invokes a conspiracy by powerful people or organizations, often without credible evidence. Conspiracy theories often involve claims of secret plots, coverups, or the manipulation of information by influential groups. Here are some examples of conspiracy theories: 1. "The moon landing was faked by the U.S. government to win the space race." 2. "The COVID-19 pandemic was planned and orchestrated by pharmaceutical companies to profit from vaccine sales." 3. "Climate change is a hoax perpetrated by scientists and politicians to gain funding and control the population." And here are some examples of statements that are not conspiracy theories: 4. "The Watergate scandal involved a cover-up of illegal activities by the Nixon administration." 5. "The tobacco industry concealed the harmful effects of smoking for many years." 6. "Corporate lobbying influences political decisions in favor of special interests." For each statement provided, respond with either "Conspiracy theory" or "Not conspiracy theory".
Prompt 3	“Analyze the following statement using a chain-of-thought approach to determine if it describes a conspiracy theory that the author believes in. Look for indicators such as personal endorsement, certainty in language, and elements that typically characterize conspiracy theories. Briefly note your observations. Present your conclusions in the following JSON format: <pre>{ "Chain_of_Thought": { "Belief_Assessment_Reasoning": "Brief justification", "Belief_Assessment": { "Belief": "Yes/No" } } }</pre>

Table S4.

GPT-4 prompts used to classify statements of belief as conspiracy theories

Pair	Agree on Yes	Agree on No	Disagree (Yes/No)	Disagree (No/Yes)
Prompt 1 (Raw Text) & Prompt 1 (GPT Summaries)	79.9%	12.7%	3.5%	3.9%
Prompt 1 (Raw Text) & Prompt 2 (GPT Summaries)	79.2%	11.0%	4.8%	5.0%
Prompt 1 (Raw Text) & Prompt 3 (Raw Text)	75.1%	12.4%	8.3%	4.2%
Prompt 1 (GPT Summaries) & Prompt 2 (GPT Summaries)	78.3%	9.8%	6.1%	5.9%
Prompt 1 (GPT Summaries) & Prompt 3 (Raw Text)	73.9%	10.9%	9.9%	5.3%
Prompt 2 (GPT Summaries) & Prompt 3 (Raw Text)	73.4%	9.5%	10.7%	6.3%

Table S5.

Agreement and disagreement across GPT-4 prompts classifying each conspiracy statement (Sample 1)

5

Pair	Agree on Yes	Agree on No	Disagree (Yes/No)	Disagree (No/Yes)
Prompt 1 (Raw Text) & Prompt 1 (GPT Summaries)	70.3%	21.3%	4.1%	4.2%
Prompt 1 (Raw Text) & Prompt 2 (GPT Summaries)	70.7%	17.0%	3.7%	8.6%
Prompt 1 (Raw Text) & Prompt 3 (Raw Text)	60.3%	23.2%	14.2%	2.4%
Prompt 1 (GPT Summaries) & Prompt 2 (GPT Summaries)	70.3%	16.4%	4.2%	9.0%
Prompt 1 (GPT Summaries) & Prompt 3 (Raw Text)	59.0%	21.8%	15.6%	3.6%
Prompt 2 (GPT Summaries) & Prompt 3 (Raw Text)	59.4%	17.5%	19.9%	3.2%

Table S6.

Agreement and disagreement across GPT-4 prompts classifying each conspiracy statement (Sample 2)

Variable	Beta	95% CI [†]	p-value
(Intercept)	82.7	80.5, 84.9	<0.001
Experimental Condition			
Control	0.000	—	
Treatment	-16.6	-19.5, -13.7	<0.001
Pre-treatment Belief	0.942	0.793, 1.09	<0.001
Experimental Condition * Pre-treatment Belief			
Treatment * Pre-treatment Belief	0.043	-0.156, 0.242	0.67

[†] CI = Confidence Interval

R² = 0.401; Adjusted R² = 0.398; Sigma = 20.2; Statistic = 169; p-value = <0.001; df = 3; Log-likelihood = -3,364; AIC = 6,738; BIC = 6,761; Deviance = 307,889; Residual df = 757; No. Obs. = 761

Table S7.

The effect of AI-human conversations on focal conspiracy beliefs in Sample 1

Variable	Beta	95% CI [†]	p-value
(Intercept)	78.3	76.2, 80.3	<0.001
Experimental Condition			
Control	0.000	—	
Treatment	-12.5	-14.9, -10.0	<0.001
Pre-treatment Belief	0.920	0.785, 1.05	<0.001
Experimental Condition * Pre-treatment Belief			
Treatment * Pre-treatment Belief	0.013	-0.146, 0.172	0.87

[†] CI = Confidence Interval

R² = 0.357; Adjusted R² = 0.356; Sigma = 21.0; Statistic = 249; p-value = <0.001; df = 3; Log-likelihood = -6,017; AIC = 12,044; BIC = 12,070; Deviance = 591,221; Residual df = 1,345; No. Obs. = 1,349

Table S8.

The effect of AI-human conversations on focal conspiracy beliefs in Sample 2

Characteristic	Beta	95% CI [†]	p-value
(Intercept)	84.0	81.5, 86.6	<0.001
ExperimentalCondition			0.8
Control	0.000	—	
Treatment	-0.471	-3.78, 2.84	
Time			0.004
Before Conversation	0.000	—	
After Conversation	-1.00	-3.35, 1.35	
10-Day Follow-Up	-2.67	-5.19, -0.151	
2-Month Follow-Up	-4.73	-7.40, -2.06	
ExperimentalCondition * Time			<0.001
Treatment * After Conversation	-16.7	-19.8, -13.6	
Treatment * 10-Day Follow-Up	-14.5	-17.8, -11.2	
Treatment * 2-Month Follow-Up	-13.0	-16.5, -9.51	
ResponseId.sd__(Intercept)	17.5		
Residual.sd__Observation	15.2		

[†] CI = Confidence Interval

No. Obs. = 2,693; Sigma = 15.2; Log-likelihood = -11,793; AIC = 23,606; BIC = 23,665; REMLcrit = 23,586; Residual df = 2,683

Table S9.

The effect of AI-human conversations on focal conspiracy over time

Characteristic	Beta	95% CI ¹	p-value
(Intercept)	38.5	36.2, 40.9	<0.001
ExperimentalCondition			0.3
Control	0.000	—	
Treatment	-1.64	-4.72, 1.43	
Time			0.3
Before Conversation	0.000	—	
After Conversation	0.839	-0.169, 1.85	
10-Day Follow-Up	0.917	-0.164, 2.00	
2-Month Follow-Up	0.822	-0.325, 1.97	
ExperimentalCondition * Time			<0.001
Treatment * After Conversation	-3.89	-5.21, -2.57	
Treatment * 10-Day Follow-Up	-3.53	-4.95, -2.12	
Treatment * 2-Month Follow-Up	-3.28	-4.79, -1.78	
ResponseId.sd__(Intercept)	20.2		
Residual.sd__Observation	6.43		

¹ CI = Confidence Interval

No. Obs. = 2,648; Sigma = 6.43; Log-likelihood = -10,013; AIC = 20,046; BIC = 20,105; REMLcrit = 20,026; Residual df = 2,638

Table S10.

The effect of AI-human conversations on general conspiracy beliefs from the BCTI over time

5

Characteristic	Beta	95% CI ¹	p-value
(Intercept)	75.5	73.4, 77.5	<0.001
ExperimentalCondition			0.6
Control	0.000	—	
Treatment	-0.800	-3.52, 1.92	
Time			<0.001
Before Conversation	0.000	—	
After Conversation	-3.32	-5.30, -1.34	
10-Day Follow-Up	-7.12	-9.24, -5.01	
2-Month Follow-Up	-10.5	-12.8, -8.29	
ExperimentalCondition * Time			<0.001
Treatment * After Conversation	-6.07	-8.67, -3.47	
Treatment * 10-Day Follow-Up	-5.09	-7.86, -2.32	
Treatment * 2-Month Follow-Up	-4.54	-7.48, -1.61	
ResponseId.sd__(Intercept)	13.0		
Residual.sd__Observation	11.9		

¹ CI = Confidence Interval

No. Obs. = 2,329; Sigma = 11.9; Log-likelihood = -9,598; AIC = 19,216; BIC = 19,273; REMLcrit = 19,196; Residual df = 2,319

Table S11.

The effect of AI-human conversations on generical conspiracy beliefs from the BCTI that participants endorsed pre-treatment.

Name	Frequency		Representative Conspiracy	GPT-4 Summary of Entire Cluster
	S1	S2		
JFK	12.2	16.8	I believe there was a plot to kill former President John F. Kennedy. Evidence points to the fact that there was more than one gunshot fired. Possibly, there was another shooter(s) that were involved to confuse authorities during their investigations. It's possible that former Vice President Lyndon B. Johnson knew that plan and had a hit ordered on Kennedy. Johnson wanted to be President and thought assassinating Kennedy was the answer. I have read articles and books that discuss this theory in length. I cannot remember where I read the articles or which books. There also have been television specials delving into inconsistencies in the original investigation of Kennedy's assassination.	Skepticism and conspiracy theories concerning the official account of President Kennedy's assassination, proposing various theories that suggest involvement by government agencies like the CIA, multiple shooters, and a broader conspiracy. Key themes include doubts about Lee Harvey Oswald acting alone, the possibility of a cover-up involving various powerful entities, and the suggestion of motives tied to JFK's policies and actions. These perspectives highlight a persistent belief in a complex plot behind Kennedy's assassination, challenging the lone gunman theory with arguments about implausible bullet trajectories and inconsistencies in evidence.
Aliens	14.0	12.1	That Area 51 could potentially be the site of alien contact. It is compelling to me because I want to know that we are not alone in the entire universe. It is also compelling because of the extra advanced technology that it would have taken to get here to Earth. The Phoenix Lights are so convincing to me. I feel like that was actually aliens. Also, the Roswell incident and the LA UFO battle lead me to believe that we might not be getting the entire truth that is out there. I do not buy some of the explanations that these are merely bad recordings.	The U.S. government, among others, is concealing evidence of extraterrestrial life and UFOs, particularly in relation to secretive sites like Area 51 and incidents like the Roswell crash. Many assertions cite declassified documents, whistleblower accounts, and personal sightings as evidence of extraterrestrial visits and government cover-ups, suggesting a significant disconnect between public information and alleged government knowledge on the matter. The statements collectively highlight a deep-seated suspicion that there is more to know about extraterrestrial existence and human-alien interactions than is officially acknowledged.
COVID-19	11.7	6.3	Covid-19 was man made and it was nothing but a simulation for the global elites to see how much they can control the masses I have seen videos of people that work in the medical field warning the masses about the global elites plan	A range of theories and beliefs concerning the origins and nature of COVID-19 and the vaccine developed in response. The assertions vary widely, from the virus being a natural occurrence to it being a lab-made bioweapon. Many statements suggest deliberate manipulation or cover-up by various governments or organizations, with some highlighting potential ulterior motives such as population control or economic gain. Concerns about the safety and efficacy of COVID-19 vaccines are prevalent, with numerous claims of adverse effects and skepticism about the vaccines' development and promotion.
9/11	7.1	5.8	9/11 was an inside job. Too many Americans benefited/profited from it for it to be a coincidence or some one-off attack. Ultimately it was a very methodical and calculated maneuver that required some degree of active participation or at least complicity from American leaders. Whatever Alex Jones said is probably what I found most compelling.	Various conspiracy theories regarding the September 11 attacks, suggesting that they were an inside job or that the US government had foreknowledge and allowed them to happen. The assertions point to perceived inconsistencies in the official account, such as the collapse of the Twin Towers and Building 7, the attack on the Pentagon, and the flight path and capabilities of the hijackers. Some theories suggest that the attacks were orchestrated to justify the invasions of Afghanistan and Iraq, to secure oil resources, or to implement the Patriot Act and enhance surveillance. Others hint at financial motives, citing insurance policies and financial anomalies related to the World Trade Center. The theories often reference controlled demolition, prior intelligence warnings, and supposed benefits to certain individuals or sectors, such as defense contractors, as part of their argumentation.
Illuminati / New World Order	2.6	6.4	World leaders destroying all governments, businesses, and capitalism in order to bring forth the New World Order agenda. This resonates with me because this isn't a conspiracy theory. It has always been a fact and we are currently witnessing the communist style	A range of conspiracy theories suggesting that various secretive groups or elite individuals exert significant, often malevolent, control over global events, economies, and governments. Theories include the Illuminati's influence on celebrities and global events,

			government, which is what will be used under the New World Order, manifest right before our eyes. Before the 20th century, there never existed in the world a welfare state. Citizens are being taxed heavily, can barely afford to eat, and can't afford to buy a house. World leaders and so-called elites gather around for various events yearly, who are un-elected, dictating how civilians everywhere should live. Being told repeatedly that you will own nothing and be happy. Climate change is not about the environment, it is about total control and world domination. These are all communistic values!	the New World Order's alleged attempts to establish a global government, and the control exerted by groups like the Freemasons or Bilderberg. Other assertions involve the manipulation of the music and entertainment industries, political systems, and economic structures by these secretive elites, suggesting they shape societal norms and policies to maintain their power and wealth. The statements often reference perceived evidence, such as symbols, policies, and the actions of high-profile individuals, to support claims of a hidden agenda aimed at manipulating public perception and global outcomes.
Malevolent Corporations	6.6	3.3	I have heard that the cure to cancer, aids, and many other illnesses have been discovered, but that the government and large pharmaceutical corporations are hiding the information/ have buried it because it would put them out of business. I don't trust big business and believe that corporations are greedy and do not care about people, so I believe that this could be true. We have so many medical and technological advancements, and so many brilliant minds have been studying these types of illnesses that it just seems unlikely that we have not made any advancements towards finding a real cure for these things.	These statements cover a range of conspiracy theories and critical views on corporate practices. They suggest that corporations engage in deceptive practices to manipulate consumer behavior, suppress technological advancements, and prioritize profits over public well-being. Theories include the manipulation of consumer products, suppression of environmentally friendly technologies, and unethical practices within the healthcare, food, and energy industries. The statements reflect a skepticism towards corporate motives, suggesting that these entities have the power and incentive to engage in activities that are not in the public interest, often with a focus on maintaining market dominance, driving consumption, or suppressing competition.
Moon Landing	3.8	4.9	I believe the moon landings were faked. It is quite clear, when you look closely at the facts with a truly open mind, that we did not (and still do not) have the technology to transport humans into space any further than earth's orbit. For example, the Van Allen radiation belts would have killed any astronauts on the way. The moon landings were filmed in a studio to raise Americans' spirits after the Space Race. Only a few people knew the truth and they were forbidden from being honest about it for reasons of national security. It has been many decades and no other country has been able to land on the moon, even though countries like China absolutely would have the funds and motivation to do so in order to showcase their own technology to the world.	This cluster of statements centers on the conspiracy theory that the Apollo moon landings were fabricated. These assertions highlight skepticism about the authenticity of the moon landing, suggesting that it was staged by the US government or NASA to win the space race against the Soviet Union, enhance national prestige, or for other geopolitical or propaganda reasons. Key points of contention include alleged inconsistencies in the lunar surface footage, such as the behavior of the American flag, the absence of stars in the sky, the quality of the photographic and video evidence, and the technology available at the time. Some claims suggest the involvement of filmmakers, specifically Stanley Kubrick, in creating the moon landing footage. Others point to the lack of subsequent manned moon missions as further evidence of the original landing's inauthenticity.
2020 Election Fraud	4.0	4.7	The one that really sticks out is the conspiracy by the deep state to steal the 2020 election from President Trump. I do not believe that Biden won, and I think the election was definitely stolen. Evidence of illegal ballot harvesting, illegal ballot drop off boxes, dead voters voting, improper signature verification, late night ballot drops for Biden, illegal immigrants voting, voting machine malfunctions, voting machine switching votes for Trump to Biden, illegal (and fake) mail in ballots, and voting after the polls closed are some of the documented examples.	The 2020 U.S. presidential election was subject to fraud and manipulation, particularly focusing on mail-in ballots, vote counting irregularities, and the alleged involvement of various domestic and foreign entities. Many assertions suggest that these alleged irregularities led to an illegitimate outcome favoring President Joe Biden over President Donald Trump, with some statements citing specific incidents and documentaries as evidence. The cluster also includes perspectives on foreign influence in U.S. elections and the perceived alignment of certain politicians with foreign interests.
Jeffrey Epstein	2.8	4.1	Jeff Epstein did not kill himself resonates with me. The elite of the world needed to silence him, and it will be coming out sooner than later all of the scum who joined him in his pedo island. It's a conspiracy there is no evidence. It just makes sense that influential people would need him to be silenced.	The death of Jeffrey Epstein, suggesting that it was not a suicide but rather a murder orchestrated to protect powerful individuals connected to him. Many comments highlight the irregularities and lapses in prison security protocols at the time of his death, such as malfunctioning cameras and guard negligence. The theories suggest that Epstein had incriminating information on influential figures, which could have motivated his assassination to prevent the exposure of their involvement in criminal activities. Some

				statements also explore the idea that Epstein's death was staged or that he might still be alive, leveraging his connections to evade justice.
MLK	3.1	1.5	Martin Luther King Jr. was assassinated by law enforcement agencies under orders from political leaders who viewed him as a threat to the status quo, employing James Earl Ray as a scapegoat, to halt the social and economic advancement of African Americans.	A belief that U.S. government agencies, particularly the FBI and CIA, were involved in the assassination of Martin Luther King Jr., viewing him as a threat to the status quo due to his civil rights activism and influence. These theories often point to the government's surveillance and discreditation efforts against King, the use of James Earl Ray as a scapegoat, and a broader pattern of government opposition to civil rights movements and leaders. The statements reflect deep skepticism toward the official narrative, indicating a suspicion of a coordinated effort to silence King's push for social change.
Princess Diana	2.3	1.2	The conspiracy theory I find most compelling is that Princess Diana was killed under the direction of the royal family. One thing about conspiracy theories is that they cannot involve too many people or else someone is bound to slip. For this to work, not very many people needed to know. Also, there are so many reasons why the royal family wanted her dead considering the massive popularity and influence she would continue to hold for the foreseeable future. We have seen how much control and power the royal family has and we have also seen they will do anything to protect it. I do not know what specific evidence led me to believe the conspiracy theory. I think it was a gradual accumulation of the events through watching documentaries and reading new stories about it. After learning about how much power and influence the royal family has and the problems Diana was causing, it seemed to make sense that they would want her dead and could make it happen. Also, considering the nature of the death, it is plausible that something more was going on.	This cluster of statements revolves around the conspiracy theory that Princess Diana's death was not accidental but orchestrated. Many assertions suggest that her death was planned due to various reasons: her knowledge of royal family secrets, her relationship with Dodi Al-Fayed, and her overall popularity and influence that posed a threat to the monarchy. Theories include involvement by Prince Charles, the broader royal family, or other powerful entities, with motives ranging from allowing Charles to remarry, to silencing Diana due to her outspokenness and potential revelations about the royal family. Suspicious details cited include anomalies in the crash investigation, like malfunctioning tunnel cameras, unusual levels of driver intoxication, delayed emergency response, and Diana's own premonitions about a car accident. Some theories extend to suggesting that the royal family's displeasure with Diana's actions or relationships prompted them to facilitate her death.
Highly Polarized	1.6	1.1	The education department of the US is deeply slanted to immoral curriculum in our schools and groom children to extreme left wing views. The education of children should not be a nonprofit effort by the government to manipulate the knowledge and emotions in a progressive direction. The low instances of trans genderism is fostered by schools and liberal teachers which is pulling these misguided children further off track and this catastrophe is fed by the department of education.	These statements reflect various conspiracy theories and critical perspectives regarding the actions and motivations of political parties, groups, and individuals in the United States. They suggest deliberate strategies by Republicans and Democrats to manipulate societal norms, election outcomes, and governmental structures for ideological gains or power consolidation. These theories range from efforts to reshape the judiciary and educational systems to influence over immigration policies and national identity. They underscore a deep polarization and mistrust in the political discourse, where each side accuses the other of undermining democracy, civil liberties, and the nation's foundational values.

Table S12.

Descriptions and representative conspiracy statements for each DBSCAN cluster

Variable	Beta	95% CI [†]	p-value
(Intercept)	79	76, 82	<0.001
Experimental Condition			<0.001
Control	0.00	—	
Treatment	-13	-16, -9.2	
Conspiracy Cluster			0.99
Not Classified (29.00%)	0.00	—	
JFK (15.14%)	2.3	-2.8, 7.5	
Aliens (12.79%)	1.8	-3.2, 6.9	
COVID-19 (8.25%)	3.2	-2.7, 9.2	
September 11th (6.30%)	1.4	-5.4, 8.2	
Illuminati / New World Order (4.98%)	0.53	-7.2, 8.2	
Malevolent Corporations (4.49%)	3.0	-5.0, 11	
Moon Landing (4.49%)	-1.4	-9.0, 6.2	
2020 Election Fraud (4.44%)	4.7	-2.6, 12	
Jeffrey Epstein (3.66%)	0.66	-8.2, 9.6	
MLK (2.05%)	3.2	-9.3, 16	
Princess Diana (1.61%)	2.0	-13, 17	
Highly Polarized (1.27%)	5.3	-11, 22	
Pre-treatment Belief	0.91	0.85, 0.97	<0.001
Experimental Condition * Conspiracy Cluster			0.21
Treatment * JFK (15.14%)	-4.2	-10, 1.9	
Treatment * Aliens (12.79%)	-3.4	-9.7, 2.9	
Treatment * COVID-19 (8.25%)	0.97	-6.4, 8.4	
Treatment * September 11th (6.30%)	-5.7	-14, 2.6	
Treatment * Illuminati / New World Order (4.98%)	4.1	-5.2, 13	
Treatment * Malevolent Corporations (4.49%)	-5.0	-15, 4.8	
Treatment * Moon Landing (4.49%)	-6.1	-16, 3.4	
Treatment * 2020 Election Fraud (4.44%)	2.2	-7.2, 12	
Treatment * Jeffrey Epstein (3.66%)	5.3	-5.5, 16	
Treatment * MLK (2.05%)	-9.7	-25, 5.1	
Treatment * Princess Diana (1.61%)	-18	-34, -0.64	
Treatment * Highly Polarized (1.27%)	8.4	-11, 28	

[†] CI = Confidence Interval

R² = 0.395; Adjusted R² = 0.387; Sigma = 20.6; Statistic = 49.4; p-value = <0.001; df = 26; Log-likelihood = -8,867; AIC = 17,789; BIC = 17,946; Deviance = 836,963; Residual df = 1,971; No. Obs. = 1,998

Table S13.

The effect of AI - human conversations on focal conspiracy beliefs by type of conspiracy theory (based on a density-based spatial clustering algorithm)

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	80.117	0.803	99.754	0.0000 ***
	ExperimentalConditionTreatment	-14.354	0.983	-14.605	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(Pre_Belief_Specific_center):ExperimentalConditionControl	1.005	1.010	313.381	0.0000 ***
	s(Pre_Belief_Specific_center):ExperimentalConditionTreatment	2.589	3.191	203.262	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '.' < 0.05

Adjusted R-squared: 0.371, Deviance explained 0.373

-REML : 9033.679, Scale est: 434.274, N: 2028

Table S14.

The effect of AI - human conversations as a function of initial focal conspiracy belief in a generalized additive model (corresponding to Figure 3a)

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Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	77.983	1.030	75.744	0.0000 ***
	ExperimentalConditionTreatment	-12.081	1.226	-9.857	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(Pre_Belief_Specific_center)	1.682	2.081	209.183	0.0000 ***
	s(Importance):ExperimentalConditionControl	1.001	1.001	6.581	0.0104 *
	s(Importance):ExperimentalConditionTreatment	1.001	1.003	60.834	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.386, Deviance explained 0.388

-REML : 5981.310, Scale est: 418.992, N: 1349

Table S15.

The effect of AI - human conversations as a function of focal conspiracy importance in a generalized additive model (corresponding to Figure 3b)

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	82.548	1.192	69.226	0.0000 ***
	ExperimentalConditionTreatment	-16.653	1.541	-10.807	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(Pre_Belief_Specific_center)	1.787	2.212	134.122	0.0000 ***
	s(Pre_Belief_General_center):ExperimentalConditionControl	1.002	1.005	1.606	0.2053
	s(Pre_Belief_General_center):ExperimentalConditionTreatment	2.585	3.245	3.779	0.0099 **

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '.' < 0.05

Adjusted R-squared: 0.399, Deviance explained 0.404

-REML : 3210.156, Scale est: 411.567, N: 726

Table S16.

The effect of AI - human conversations as a function of belief in non-focal conspiracies (BCTI scores) in a generalized additive model (corresponding to Figure 3c)

Characteristic	Beta	95% CI [†]	p-value
(Intercept)	83.0	74.2, 91.9	<0.001
Experimental_Condition			<0.001
<i>Control</i>	0.000	—	
<i>Active</i>	-21.3	-32.8, -9.81	
Pre_Belief_Specific_center	0.896	0.729, 1.06	<0.001
Generative_AI_Familiarity	0.395	-2.52, 3.31	0.79
Generative_AI_Usage	0.088	-3.15, 3.32	0.96
Generative_AI_Trust	-0.155	-2.87, 2.57	0.91
Religiosity	0.240	-2.57, 3.05	0.87
Partisanship	1.60	-1.02, 4.22	0.23
Extremism	0.010	-2.98, 3.00	>0.99
Years_of_Age	-0.172	-2.94, 2.59	0.90
Conspiracy_Type			>0.99
<i>Not Classified (29.00%)</i>	0.000	—	
<i>JFK (15.14%)</i>	1.09	-7.60, 9.78	
<i>Aliens (12.79%)</i>	1.75	-5.75, 9.25	
<i>COVID-19 (8.25%)</i>	0.836	-7.30, 8.98	
<i>September 11th (6.30%)</i>	2.14	-7.02, 11.3	
<i>Illuminati / New World Order (4.98%)</i>	-1.34	-14.0, 11.3	
<i>Malevolent Corporations (4.49%)</i>	2.00	-8.56, 12.6	
<i>Moon Landing (4.49%)</i>	-4.82	-19.5, 9.82	
<i>2020 Election Fraud (4.44%)</i>	4.50	-6.10, 15.1	
<i>Jeffrey Epstein (3.66%)</i>	5.27	-9.29, 19.8	
<i>MLK (2.05%)</i>	4.70	-10.9, 20.3	
<i>Princess Diana (1.61%)</i>	1.32	-22.8, 25.4	
<i>Highly Polarized (1.27%)</i>	6.59	-21.8, 34.9	
Education_Category			0.95
<i>SomeCollege</i>	0.000	—	
<i>Associate</i>	0.842	-7.02, 8.70	
<i>Bachelors</i>	-0.396	-6.63, 5.84	
<i>HighSchool</i>	-1.92	-10.6, 6.77	
<i>JD/MD</i>	2.83	-18.5, 24.1	
<i>LessThanHighSchool</i>	-6.85	-35.0, 21.3	
<i>Masters</i>	1.48	-7.58, 10.5	
<i>PhD</i>	-10.7	-29.2, 7.85	
Race_Category			>0.99
<i>White</i>	0.000	—	
<i>Asian</i>	0.275	-12.9, 13.5	
<i>Black</i>	0.482	-6.80, 7.77	
<i>Other</i>	-1.90	-19.7, 15.9	
Gender_Category			0.55
<i>female</i>	0.000	—	
<i>male</i>	-2.26	-7.37, 2.86	
<i>other</i>	-8.14	-29.2, 12.9	

Experimental_Condition * Pre_Belief_Specific_center		0.85
Active * Pre_Belief_Specific_center	-0.021	-0.238, 0.196
Experimental_Condition * Generative_AI_Familiarity		0.42
Active * Generative_AI_Familiarity	1.52	-2.21, 5.25
Experimental_Condition * Generative_AI_Usage		0.38
Active * Generative_AI_Usage	1.85	-2.28, 5.97
Experimental_Condition * Generative_AI_Trust		0.004
Active * Generative_AI_Trust	-5.26	-8.78, -1.73
Experimental_Condition * Religiosity		0.74
Active * Religiosity	-0.602	-4.15, 2.94
Experimental_Condition * Partisanship		0.76
Active * Partisanship	0.563	-3.03, 4.15
Experimental_Condition * Extremism		0.44
Active * Extremism	1.57	-2.41, 5.56
Experimental_Condition * Years_of_Age		0.003
Active * Years_of_Age	5.51	1.93, 9.10
Experimental_Condition * Conspiracy_Type		0.47
Active * JFK (15.14%)	-9.48	-20.3, 1.31
Active * Aliens (12.79%)	-4.83	-14.6, 4.94
Active * COVID-19 (8.25%)	-0.533	-11.1, 10.0
Active * September 11th (6.30%)	-12.6	-24.8, -0.439
Active * Illuminati / New World Order (4.98%)	10.7	-8.39, 29.7
Active * Malevolent Corporations (4.49%)	-5.00	-18.4, 8.43
Active * Moon Landing (4.49%)	2.59	-15.1, 20.2
Active * 2020 Election Fraud (4.44%)	2.54	-13.5, 18.5
Active * Jeffrey Epstein (3.66%)	9.86	-9.73, 29.5
Active * MLK (2.05%)	-5.96	-25.4, 13.5
Active * Princess Diana (1.61%)	-0.077	-26.9, 26.7
Active * Highly Polarized (1.27%)	-1.45	-32.7, 29.8
Experimental_Condition * Education_Category		0.82
Active * Associate	2.97	-7.62, 13.6
Active * Bachelors	2.50	-5.52, 10.5
Active * HighSchool	5.43	-6.07, 16.9
Active * JD/MD	4.94	-20.7, 30.6
Active * LessThanHighSchool	-21.6	-56.1, 13.0
Active * Masters	-0.276	-11.7, 11.1
Active * PhD	9.99	-12.7, 32.7
Experimental_Condition * Race_Category		0.94
Active * Asian	-4.44	-20.0, 11.1
Active * Black	-1.49	-11.0, 7.99
Active * Other	-1.69	-30.9, 27.6
Experimental_Condition * Gender_Category		0.13
Active * male	6.13	-0.408, 12.7
Active * other	21.2	-23.8, 66.2

[†] CI = Confidence Interval

R² = 0.497; Adjusted R² = 0.446; Sigma = 19.4; Statistic = 9.82; p-value = <0.001; df = 65; Log-likelihood = -3,092; AIC = 6,318; BIC = 6,624; Deviance = 243,905; Residual df = 647; No. Obs. = 713

Table S17.

The effect of AI - human conversations on focal conspiracy beliefs by all covariates in Sample 1

Characteristic	Beta	95% CI [†]	p-value
(Intercept)	77.3	71.1, 83.5	<0.001
Experimental_Condition			0.003
Control	0.000	—	
Active	-11.2	-18.5, -3.87	
Pre_Belief_Specific_centered	0.935	0.788, 1.08	<0.001
Generative_AI_Familiarity	0.593	-1.94, 3.13	0.65
Generative_AI_Usage	0.265	-2.59, 3.12	0.86
Generative_AI_Trust	0.170	-2.31, 2.65	0.89
Religiosity	-0.046	-2.47, 2.38	0.97
Partisanship	1.04	-1.57, 3.65	0.43
Extremism	0.541	-1.79, 2.87	0.65
Years_of_Age	0.450	-2.10, 3.00	0.73
dbscan_cluster			>0.99
Not Classified (29.00%)	0.000	—	
JFK (15.14%)	2.85	-3.90, 9.60	
Aliens (12.79%)	1.19	-5.79, 8.17	
COVID-19 (8.25%)	2.51	-6.63, 11.7	
September 11th (6.30%)	-0.504	-10.7, 9.67	
Illuminati / New World Order (4.98%)	-0.011	-9.83, 9.81	
Malevolent Corporations (4.49%)	3.49	-9.52, 16.5	
Moon Landing (4.49%)	-1.33	-10.7, 7.98	
2020 Election Fraud (4.44%)	3.80	-6.77, 14.4	
Jeffrey Epstein (3.66%)	-1.24	-12.7, 10.2	
MLK (2.05%)	0.524	-20.2, 21.3	
Princess Diana (1.61%)	1.58	-17.5, 20.6	
Highly Polarized (1.27%)	4.32	-16.7, 25.3	
Education_Category			0.88
SomeCollege	0.000	—	
Associate	0.218	-7.03, 7.47	
Bachelors	-0.843	-6.79, 5.11	
HighSchool	1.27	-6.36, 8.90	
JD/MD	-4.71	-19.9, 10.5	
LessThanHighSchool	-1.25	-25.4, 22.9	
Masters	0.445	-7.47, 8.36	
PhD	-18.2	-42.2, 5.87	
Race_Category			0.74
White	0.000	—	
Asian	3.12	-6.02, 12.3	
Black	3.29	-3.54, 10.1	
Other	-1.02	-15.8, 13.8	
Gender_Category			0.98
female	0.000	—	
male	0.252	-4.16, 4.67	
other	1.80	-15.4, 19.0	

Experimental_Condition * Pre_Belief_Specific_centered		0.32
Active * Pre_Belief_Specific_centered	-0.086	-0.258, 0.085
Experimental_Condition * Generative_AI_Familiarity		0.54
Active * Generative_AI_Familiarity	-0.932	-3.91, 2.05
Experimental_Condition * Generative_AI_Usage		0.45
Active * Generative_AI_Usage	1.30	-2.07, 4.67
Experimental_Condition * Generative_AI_Trust		0.025
Active * Generative_AI_Trust	-3.36	-6.29, -0.425
Experimental_Condition * Religiosity		0.99
Active * Religiosity	0.019	-2.84, 2.88
Experimental_Condition * Partisanship		0.087
Active * Partisanship	2.69	-0.395, 5.77
Experimental_Condition * Extremism		0.35
Active * Extremism	1.32	-1.43, 4.06
Experimental_Condition * Years_of_Age		0.43
Active * Years_of_Age	1.20	-1.76, 4.16
Experimental_Condition * dbscan_cluster		0.49
Active * JFK (15.14%)	-3.88	-11.8, 4.05
Active * Aliens (12.79%)	1.09	-7.31, 9.49
Active * COVID-19 (8.25%)	-0.795	-11.7, 10.1
Active * September 11th (6.30%)	-0.131	-12.0, 11.7
Active * Illuminati / New World Order (4.98%)	2.55	-8.87, 14.0
Active * Malevolent Corporations (4.49%)	-4.95	-20.0, 10.1
Active * Moon Landing (4.49%)	-7.80	-19.4, 3.81
Active * 2020 Election Fraud (4.44%)	-4.70	-17.3, 7.93
Active * Jeffrey Epstein (3.66%)	7.32	-6.10, 20.7
Active * MLK (2.05%)	-5.27	-28.6, 18.1
Active * Princess Diana (1.61%)	-26.7	-49.5, -3.95
Active * Highly Polarized (1.27%)	3.95	-21.1, 29.0
Experimental_Condition * Education_Category		0.88
Active * Associate	0.259	-8.55, 9.07
Active * Bachelors	-0.502	-7.52, 6.51
Active * HighSchool	2.07	-7.09, 11.2
Active * JD/MD	7.06	-11.7, 25.8
Active * LessThanHighSchool	11.5	-17.3, 40.3
Active * Masters	-4.02	-13.3, 5.26
Active * PhD	7.27	-18.3, 32.9
Experimental_Condition * Race_Category		0.42
Active * Asian	0.015	-10.9, 11.0
Active * Black	-6.45	-14.5, 1.59
Active * Other	4.83	-13.2, 22.9
Experimental_Condition * Gender_Category		0.70
Active * male	0.366	-4.90, 5.63
Active * other	10.7	-14.3, 35.8

[†] CI = Confidence Interval

R² = 0.430; Adjusted R² = 0.399; Sigma = 20.4; Statistic = 13.9; p-value = <0.001; df = 65; Log-likelihood = -5,564; AIC = 11,263; BIC = 11,607; Deviance = 496,247; Residual df = 1,197; No. Obs. = 1,263

Table S18.

The effect of AI - human conversations on focal conspiracy beliefs by all covariates in Sample 18

Variable	Beta	95% CI [†]	p-value
(Intercept)	2.56	2.51, 2.62	<0.001
Experimental Condition			
Control	0.000	—	
Treatment	-0.219	-0.286, -0.152	<0.001

[†] CI = Confidence Interval

R² = 0.030; Adjusted R² = 0.029; Sigma = 0.572; Statistic = 41.2; p-value = <0.001; df = 1; Log-likelihood = -1,160; AIC = 2,327; BIC = 2,342; Deviance = 441; Residual df = 1,348; No. Obs. = 1,350

Table S19.
The effect of AI - human conversations on reaction to conspiracy posters on social media (response scale = 1 to 3).

Variable	Beta	95% CI [†]	p-value
(Intercept)	3.81	3.72, 3.89	<0.001
Experimental Condition			
Control	0.000	—	
Treatment	-0.360	-0.459, -0.261	<0.001

[†] CI = Confidence Interval
R² = 0.036; Adjusted R² = 0.036; Sigma = 0.846; Statistic = 51.0; p-value = <0.001; df = 1; Log-likelihood = -1,691; AIC = 3,388; BIC = 3,404; Deviance = 966; Residual df = 1,350; No. Obs. = 1,352

Table S20.

The effect of AI - human conversations on reaction to discussions with focal conspiracy believers (response scale = 1 to 5).

Variable	Beta	95% CI [†]	p-value
(Intercept)	2.47	2.31, 2.62	<0.001
ExperimentalCondition			
Control	0.000	—	
Treatment	-0.146	-0.332, 0.040	0.12

[†] CI = Confidence Interval

R² = 0.003; Adjusted R² = 0.002; Sigma = 1.23; Statistic = 2.39; p-value = 0.12; df = 1; Log-likelihood = -1,263; AIC = 2,531; BIC = 2,545; Deviance = 1,177; Residual df = 774; No. Obs. = 776

Table S21.

The effect of AI - human conversations on willingness to join protests supporting the focal conspiracy (response scale = 1 to 5).